method and the PZI concept are described in detail. The PZI analysis is made for cracks in a specific pressure vessel and the results are compared with those obtained by formulas based on yield load or limit load solutions.

LINEAR-ELASTIC SOLUTIONS

The following stress intensity factor (K) formulas are taken directly from **pages 33.6, 35.1, and 36.1**. For through-wall cracks of length, $2a$, in pressure vessels of mean radius, R, and wall thickness, t, subjected to an internal pressure, p, K values are expressed in the following form using a function of a single, dimensionless geometric shell parameter λ.

$$K = \sigma\sqrt{\pi a}F(\lambda) \tag{H1}$$

where the parameter λ is defined as

$$\lambda = \frac{a}{\sqrt{Rt}} \tag{H2}$$

and the definition of σ and the expression of $F(\lambda)$ are given below for each crack configuration.

a. For a longitudinal through-wall crack in a cylinder,

$$\begin{aligned} \sigma &= pR/t \\ F(\lambda) &= \left(1 + 1.25\lambda^2\right)^{1/2} \quad 0 \le \lambda \le 1 \\ &= 0.6 + 0.9\lambda \qquad\quad\; 1 \le \lambda \le 5 \end{aligned} \tag{H3}$$

b. For a circumferential through-wall crack in a cylinder,

$$\begin{aligned} \sigma &= pR/(2t) \\ F(\lambda) &= \left(1 + 0.3225\lambda^2\right)^{1/2} \quad 0 \le \lambda \le 1 \\ &= 0.9 + 0.25\lambda \qquad\quad\; 1 \le \lambda \le 5 \end{aligned} \tag{H4}$$

c. For a circumferential through-wall crack in a spherical shell,

$$\begin{aligned} \sigma &= pR/(2t) \\ F(\lambda) &= \left(1 + 1.41\ \lambda^2 + 0.04\ \lambda^3\right)^{1/2} \quad 0 \le \lambda \le 3 \end{aligned} \tag{H5}$$

These formulas are presented here for subsequent use in the plastic zone instability analysis.

PLASTIC ZONE SIZE (r_Y) CORRECTION METHOD

A localized yielding near the crack tip begins immediately with the load application because of the presence of stress singularity and causes deviation from the behavior based on the purely elastic analysis. Therefore, rigorously speaking, there is no linear portion in the load-displacement relation for a cracked body. Interestingly, as shown later, this nonlinearity manifests itself as the linear portion of the material J (or $\mathcal{G}$)-resistance curve prior to the actual extension of the crack.

It is the plastic zone size (r_Y) correction method that is commonly used to account for the effect of such local yielding. The plastic zone size r_Y given in the form of **Eq. (H7)** with $\alpha = 2$, was initially pointed out by Paris in 1957 **(Paris, 1957)**, and subsequently theoretical grounds for the use of the r_Y-corrected effective crack size in LEFM analysis was provided by Irwin and Koskinen **(Irwin, 1963)**, The method was developed for evaluating material toughness in small-scale yielding conditions where the yielding zone near the crack tip is well contained within the surrounding elastic field.

In this analysis model, the actual crack size, a, in the elastic solution is replaced by the r_Y-corrected effective size; that is, the effective elastic solution is used. The effective crack size, a_{eff}, is given by

$$a_{eff} = a + r_Y \tag{H6}$$

where

$$r_Y = \frac{1}{\alpha\pi}\left(\frac{K}{\sigma_Y}\right)^2 \tag{H7}$$

σ_Y is the yield strength of the material; the constant α is customarily taken as $\alpha = 2$ (plane stress) or $\alpha = 6$ (plane strain) depending on the constraining conditions (triaxiality) near the crack tip. It is sometimes convenient to take α as a geometry-dependent constant to extend the range of applicability of the method **(Tada, 1972a)**.

Because K on the right-hand side of **Eq. (H7)** is a function of crack size, to obtain the plastic zone size adjustment, r_Y, in a consistent manner, an iterative procedure or a graphical method is required. When K is given in a function form, as is the case of the shell crack problems of interest, the graphical method is conveniently used on the principle of "a single curve and straight lines."

The crack size is represented by the dimensionless parameter λ defined by **Eq. (H2)**. For convenience, writing $\lambda = a/b$ with $b = \sqrt{Rt}$, the following are equivalent to **Eqs. (H6) and (H7)**, respectively,

$$\lambda_{eff} = \lambda + \lambda_Y \tag{H8}$$

where

$$\lambda_Y = \frac{r_Y}{b} = \frac{1}{\alpha\pi b}\left(\frac{K}{\sigma_Y}\right)^2 \tag{H9}$$

Substituting **Eq. (H1) into Eq. (H9)** for K,

$$\lambda_Y = \frac{1}{\alpha}\cdot s^2 \cdot Y(\lambda_{eff}) \tag{H10}$$

where

$$\begin{aligned} s &= \sigma/\sigma_Y \\ Y(\lambda) &= \lambda\{F(\lambda)\}^2 \end{aligned} \tag{H11}$$

For a graphical solution for the unknown, λ_{eff}, it may be more convenient to write **Eq. (H10)** in the following form in terms of λ_{eff} eliminating λ_Y:

$$\lambda_{eff} - \lambda = \frac{1}{\alpha} \cdot s^2 \cdot Y(\lambda_{eff})$$

or further,

$$Y(\lambda_{eff}) = \alpha \cdot \frac{1}{s^2}(\lambda_{eff} - \lambda) \tag{H12}$$

The form of **Eq. (H12)** allows a simple graphical solution for λ_{eff} for any specified crack size parameter $\lambda = \lambda_0$ and various load levels represented by $s = \sigma/\sigma_Y$. That is, λ_{eff} is determined from the intersection points of the curve

$$y = Y(\lambda) \tag{H13}$$

and a series of straight lines

$$y = g(\lambda; \lambda_0, s) = \alpha \cdot \frac{1}{s^2}(\lambda - \lambda_0) \tag{H14}$$

The method is schematically presented in **Fig. H.1**. The λ_{eff} illustrated in the figure is $\lambda_{eff} = \lambda_{eff}(\lambda_0, s_2)$, corresponding to the given crack size λ_0 and the load level $s = s_2$.

PLASTIC ZONE INSTABILITY CONCEPT

Before discussing this concept, let us make some useful observations regarding **Fig. H.1**. From **Fig. H.1** we see that the plastic zone size, $r_Y = b\lambda_Y = \sqrt{Rt}\lambda_Y$, increases as the load level, $s = \sigma/\sigma_Y$, increases until the straight line given by **Eq. (H14)** becomes the tangent to the curve $y = Y(\lambda)$, **Eq. (H13)**, at $s = s_{\tan}$. That is, the solution of the iterative equation, **Eq. (H7)**, does not converge beyond this load level, $s > s_{\tan}$. The values of λ_Y and λ_{eff} corresponding to $s = s_{\tan}$ are also shown in **Fig. H.1** as $(\lambda_Y)_{\tan}$ and $(\lambda_{eff})_{\tan}$, respectively, along with the corresponding slope $(\alpha/s^2)_{\tan}$ of **Eq. (H14)**. Note that, although the point of tangency, A, may not be determined very accurately, the slope $\alpha/s^2_{\tan}$ is determined accurately. It is interesting to note also that the point of tangency A is unique for a given crack size λ_0 irrespective of the value of constant α. In other words, although $s_{\tan}$ depends on the value of α, the slope of the tangent, $(\alpha/s^2)_{\tan} = \alpha/s^2_{\tan}$, and the corresponding value of $(\lambda_Y)_{\tan}$ are uniquely determined for each given crack size λ_0.

Also noted is that **Fig. H.1** is an alternative form of **Fig. H.2**, which was used by **Vazquez (1971)** in his analysis of the plastic zone instability (PZI). **Figure H.1** was chosen for simplicity of graphical solution, where the load parameter $s = \sigma/\sigma_Y$ is separated from the function (λ_Y) to permit the use of a single curve and a series of straight lines instead of a series of curves and straight lines used in the method of **Fig. H.2**. However, since **Eq. (H12) and Figure H.1** are not designed for discussing the PZI concept, the alternative form, **Eq. (H17)**, and the corresponding graphical interpretation, **Fig. H.2**, are more convenient for directly following the discussion by Vazquez.

Eq. (H7) in terms of J(or $\mathcal{G}$) $= K^2/E$ is

$$r_Y = \frac{1}{2\pi}\frac{E}{\sigma_Y^2}J \tag{H15}$$

where E is the Young's modulus. Let us introduce the normalized form of J,

$$\overline{J} = \frac{J}{\left(\frac{\sigma_Y^2 b}{E}\right)} = \frac{J}{\left(\frac{\sigma_Y^2\sqrt{Rt}}{E}\right)} \tag{H16}$$

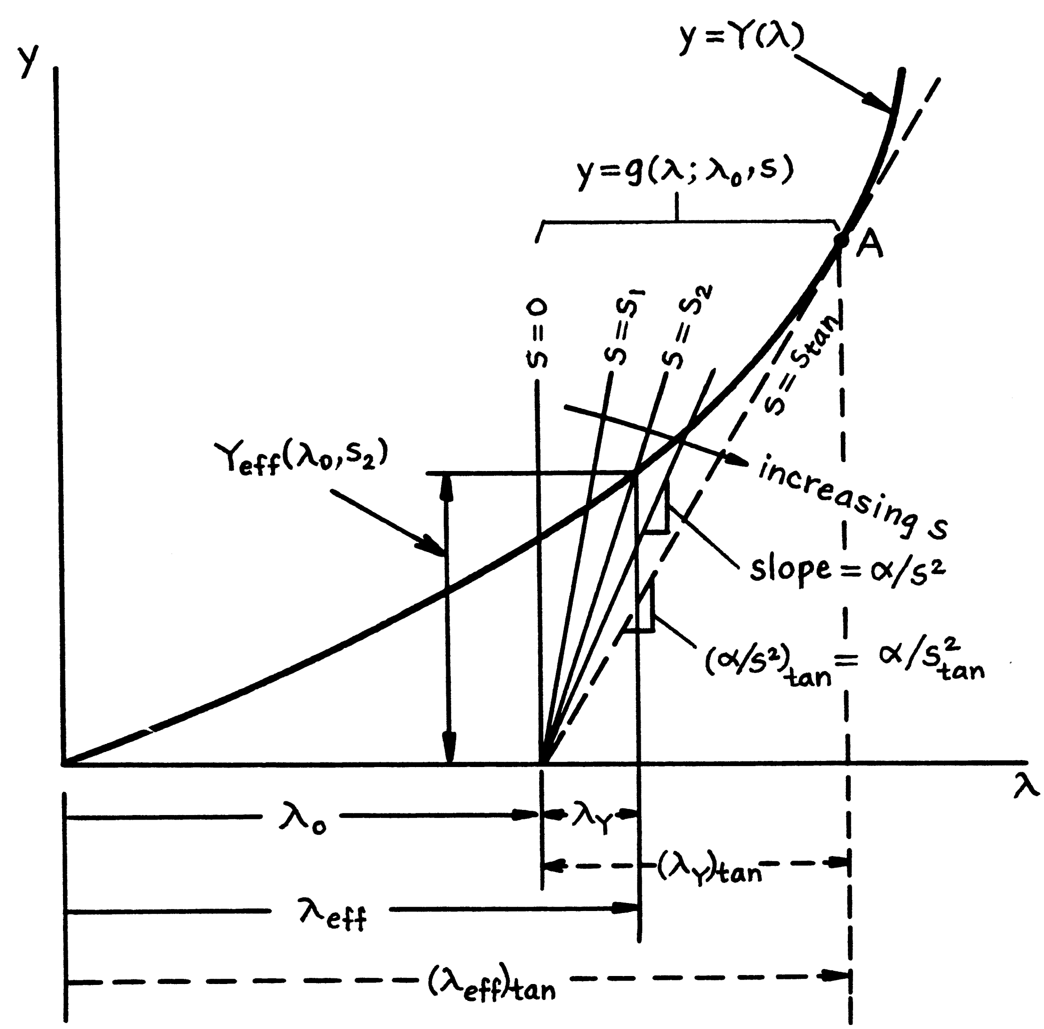


Fig. H.1. Graphical solution of Eq. (H12).

and rewrite **Eq. (H15) or (H11)** as

$$\alpha\pi\lambda_Y = \bar{J}(\lambda_{eff}, s)$$

or

$$\alpha\pi(\lambda_{eff} - \lambda) = \bar{J}(\lambda_{eff}, s) \qquad \textbf{(H17)}$$

where

$$\bar{J}(\lambda, s) = \pi s^2 Y(\lambda) \qquad \textbf{(H18)}$$

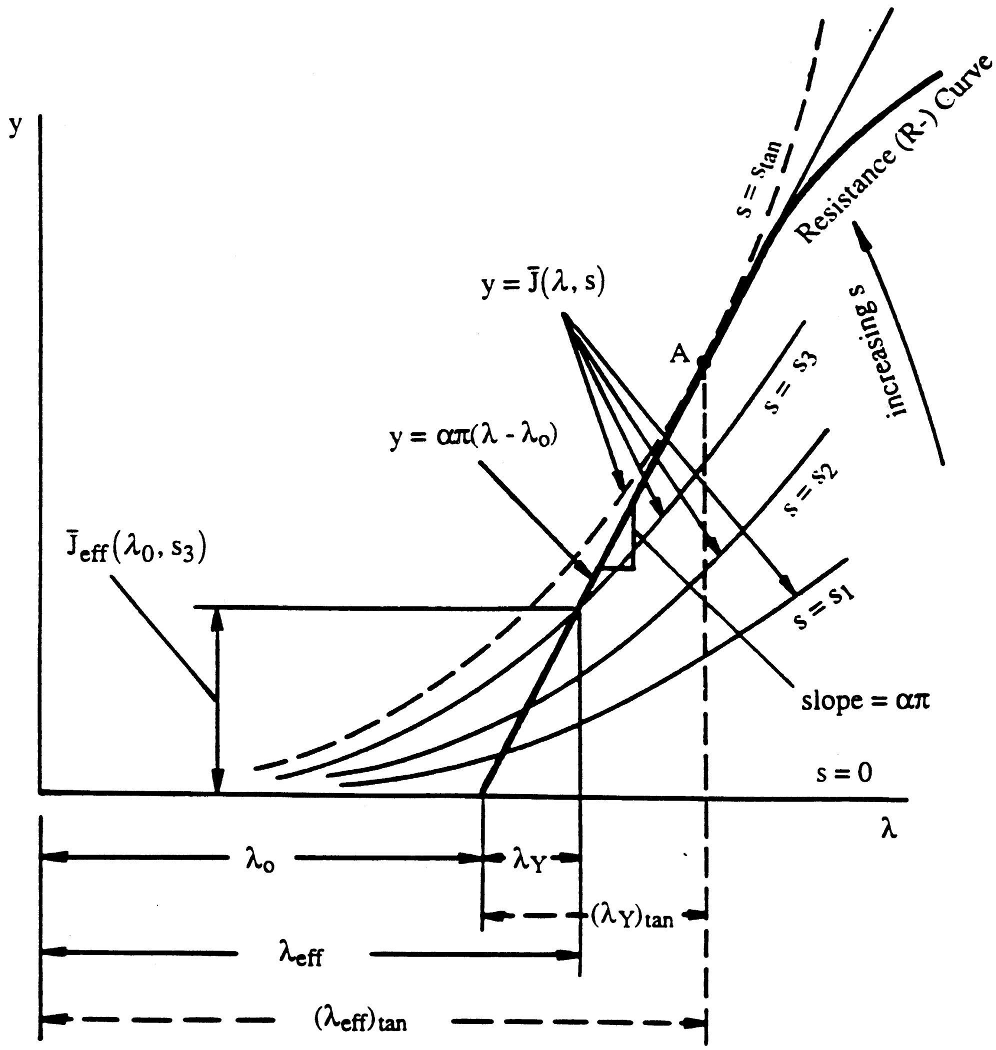


Fig. H.2. Schematic of plastic zone instability.

In **Fig. H.2**, a series of curves $y = \bar{J}(\lambda, s)$ and a straight line $y = \alpha\pi(\lambda - \lambda_0)$ are shown. The PZI concept is discussed on the basis of **Fig. H.2**, which is equivalent to **Fig. H.1**.

The possibility of extending the r_Y-corrected LEFM approach to non-small-scale yielding situations was explored and some justifications were given by **Tada (1983c, 1972a)**. **Vazquez (1971)** examined the static equilibrium between the plastic zone near the crack tip and the surrounding elastic field and proposed the concept of plastic zone instability (instability without crack growth). Detailed discussions on the mechanism and the physical interpretations of this concept are found in **Vazquez (1971)**, and some similar considerations are also found in **Tada (1972a)**. Vazquez's analysis may be summarized as follows.

Let us consider a crack extension resistance (R-Δa) curve and the interpretations of crack equilibrium and crack stability associated with it. It is noted that, as long as the crack does not extend ($\Delta a = 0$ or $\Delta a_{eff} = r_Y$), the plastic zone extension resistance $\left(R\text{-}r_Y\right)$ curve, **Eq. (H15)**, represented in **Fig. H.2** by the straight line, $y = \alpha\pi(\lambda - \lambda_0)$, constitutes the initial portion of the material resistance curve. Thus, when $F(\lambda)$ in **Eq. (H1)** is a rapidly increasing function, as in the case of through-wall cracks in shells, two distinct types of instability behaviors may be expected depending on the material's capability of absorbing plastic deformation; that is, the instability controlled by crack propagation – fracture toughness instability, and the instability controlled by plastic zone development – plastic zone instability.

It is observed from **Fig. H.2** that there is an equilibrium plastic zone size at each load level determined by the intersection point $\left(\lambda_{eff},\ \overline{J}_{eff}\right)$ between the curve $y = \overline{J}(\lambda, s)$ and the straight r_Y-line, $y = \alpha\pi(\lambda - \lambda_0)$, in the absence of crack extension. When the applied load level reaches $s = s_{\tan}$, the curve becomes tangent to the straight line at the point A and no further equilibrium plastic zone is determined. That is, for $s > s_{\tan}$ the solution for the iterative equation, **Eq. (H17)**, no longer converges. The interpretation of Vazquez is that for $s > s_{\tan}$ the plastic zone can not maintain stable equilibrium with the surrounding elastic field. If this tangency occurs before an appreciable crack extension (or deviation of the resistance curve from the straight r_Y-line) takes place, the plastic zone becomes unstable and spreads across the body. This is referred to as the plastic zone instability. The load level at instability is given by $s_{inst} = s_{\tan}$. When unstable crack propagation precedes the PZI, the fracture toughness analysis is to be followed. Note that the only material property involved in the PZI analysis is the yield strength σ_Y. Therefore, the PZI analysis alone does not determine which mode of failure occurs first.

Vazquez (1971) experimentally confirmed the prediction for existence of the PZI for a longitudinal through-wall crack in a pressurized cylinder in plane stress conditions ($\alpha = 2$) and concluded that the PZI analysis provides a conservative estimate of failure conditions.

r_Y AT PLASTIC ZONE INSTABILITY AND CONDITIONS FOR UNRESTRICTED PLASTIC FLOW

As previously observed, the values of r_Y at the occurrence of plastic instability $(r_Y)_{inst} = (r_Y)_{\tan}$ are uniquely determined by the geometry irrespective of the value of α. However, the values of r_Y at PZI do not generally represent the fully yielded conditions geometrically. The use of the LEFM analysis with the r_Y-adjusted effective crack size in such large-scale yielding conditions may seem unjustified.

Vazquez (1971) suggested that the quantity r_Y may be regarded as an index representing the development of the plastic yield zone rather than the physical plastic zone size itself in large-scale yielding conditions. Then it may be reasonable to assume the correspondence between the conditions for plastic zone instability and the attainment of the conditions for unrestricted plastic flow.

Interestingly, such correspondence is what exactly occurs in a few extreme two-dimensional crack configurations with the interpretation of r_Y as the physical size of plastic zone. Referring to **Fig. H.3**, consider a few deeply cracked bodies where the sole dimension relevant to the analysis is b, that is, the net ligament size. For the tensile configurations, **Fig. H.3**, **a and b**, the r_Y at the plastic zone instability, $(r_Y)_{\tan}$, is given by

$$\left(r_Y\right)_{\tan} = \frac{b}{2} \tag{H.19}$$

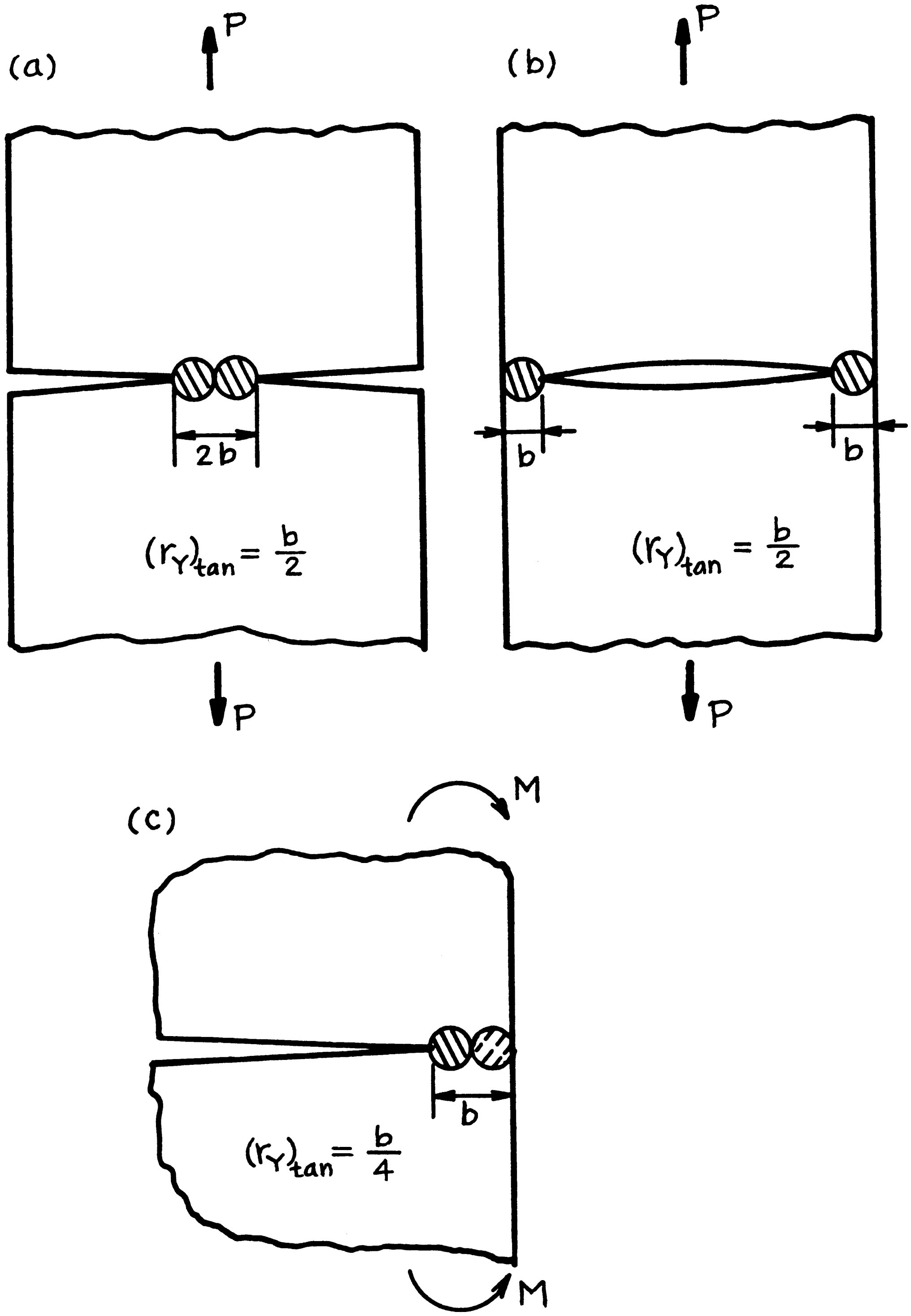


Fig. H.3. r_Y at plastic zone instability for several extreme configurations.

and for the bending configuration, **Fig. H.3c**, it is given by

$$\left(r_Y\right)_{\tan} = \frac{b}{4} \tag{H.20}$$

regardless of the value of α in **Eq. (H7)**. These conditions are readily obtained algebraically by noting that the stress intensity factors for these configurations have the following forms. For the tension configurations, **Fig. H.3**, **a and b**,

$$K = A_t \left(\frac{P}{B}\right) \frac{1}{b^{1/2}} \tag{H21}$$

and for the bending configuration, **Fig. H.3c**,

$$K = A_b \left(\frac{M}{B}\right) \frac{1}{b^{3/2}} \tag{H22}$$

where B is the thickness of the plates. The values of proportionality constants, A_t and A_b, are found on **pages 2.6, 2.1, and 9.1**. The values of A_t and A_b are immaterial for the validity of **Eqs. (H19) and (H20)**.

The use of the r_Y-adjusted effective LEFM solution may be reasonably extended to cover a wide range of elastic–plastic loading leading to the plastic zone instability. Some discussions on the use of the geometry-adjusted α are found in **Tada (1983c) and Tada (1972a)**.

APPLICATION TO A PRESSURE VESSEL

The plastic zone instability analysis was applied to determine the conditions for unrestricted plastic flow for through-wall cracks in an actual pressure vessel for a chemical reactor made of a ferritic steel. Its actual dimensions are

R = 125" (3160 mm)
t = 2.52" (64 mm) for cylindrical part
t = 1.26" (32 mm) for spherical end caps

The iterative equation, **Eq. (H7)**, was solved graphically (**Fig. H.1**) for the values of σ/σ_Y at the plastic zone instability, that is, $\sigma/\sigma_Y = s_{\tan}$, for the three crack configurations listed in the section: Linear-Elastic Solutions. The functions $F(\lambda)$ used here are **Eqs. (H3), (H4), and (H5)**. The results are presented in **Table H.1** in PZI columns. A report for the Materials Properties Council (MPC) by **Anderson (1993)** gives

$$F(\lambda) = \left(1 + 1.6\lambda^2\right)^{1/2} \tag{H23}$$

for both circumferential cracks and cracks in spheres without limits on λ. As observed from the comparison of **Eq. (H23)** with **Eqs. (H4) and (H5), Eq. (H23)** is an overestimate, especially for circumferential cracks. Therefore, the use of **Eq. (H23)** is not recommended here.

In **Table H.1**, these results are contrasted with the results in plastic collapse load (PCL) columns obtained through the following formulas of the critical stress levels, σ/σ_Y, for unrestricted plastic flow. These formulas were taken from the MPC report **(Anderson, 1993)**.

a. Longitudinal through-wall cracks in cylinders

$$\frac{\sigma}{\sigma_Y} = \left[1 + 1.6\lambda^2\right]^{-1/2} \quad \text{(H24)}$$

where $\sigma = {}^{PR}/_{t}$.

b. Circumferential through-wall cracks in cylinders

$$\frac{\sigma}{\sigma_Y} = \frac{\pi - \theta}{\pi}\left[1 + \frac{2\sin\theta\left(\cos\theta + \frac{\sin\theta}{\pi-\theta}\right)}{\pi - \theta - \frac{2\sin^2\theta}{\pi-\theta} - \frac{\sin^2\theta}{2}}\right]^{-1} \quad \text{(H25)}$$

where $\theta = {}^{a}/_{R}$, and $\sigma = pR/(2t)$. This is the form found in MPC **(Anderson, 1993)**, which is somewhat simplified to

$$\frac{\sigma}{\sigma_Y} = \frac{\pi - \theta}{\pi}\left[1 - \frac{\sin 2\theta + \frac{1-\cos 2\theta}{\pi-\theta}}{\pi - \theta + \frac{1}{2}\sin 2\theta}\right] \quad \text{(H25a)}$$

c. Through-wall cracks in spheres

$$\frac{\sigma}{\sigma_Y} = 2\left[1 + \sqrt{1 + \frac{8\lambda^2}{(\cos\theta)^2}}\right]^{-1} \quad \text{(H26)}$$

where again $\theta = {}^{a}/_{R}$ and $\sigma = pR/(2t)$.

Table H.1. Conditions for Unrestricted Plastic Flow for Through-Wall Cracks in Pressure Vessels (Plastic Zone Instability vs. Plastic Collapse Load)

	Cylinder Longitudinal Crack (${}^{\sigma}/_{\sigma_Y}$)		Cylinder Circumferential Crack (${}^{\sigma}/_{\sigma_Y}$)		Sphere Spherical Crack (${}^{\sigma}/_{\sigma_Y}$)	
λ	P.Z.I.	P.C.L.	P.Z.I.	P.C.L.	P.Z.I.	P.C.L.
0	$\sqrt{2}$	1.0	$\sqrt{2}$	1.0	$\sqrt{2}$	1.0
0.25	.900	.954	1.07	.965	.878	.899
0.50	.669	.845	.894	.932	.642	.730
0.75	.523	.726	.764	.899	.494	.596
1.0	.423	.620	.661	.866	.396	.494
1.5	.314	.446	.605	.800	.279	.365
2.0	.250	.368	.526	.737	.213	.285
2.5	.206	.302	.460	.677	.170	.229
3.0	.170	.255	.418	.619	.142	.191
	$\sigma = \frac{pR}{t}$		$\sigma = \frac{pR}{2t}$		$\sigma = \frac{pR}{2t_{sp}}$	

The actual dimensions of the pressure vessel analyzed are R = 125" (3160 mm), t=2.52" (64 mm), and t_{sp} = 1.26" (32mm).

DISCUSSIONS

Some observations can be made from the results presented in **Table H.1**.

The critical stress values σ/σ_Y for unrestricted plastic flow predicted by the plastic zone instability analysis are generally lower than those predicted by the formulas based on the plastic collapse (the yield or limit) loads **(Anderson, 1993)** except for very small cracks. That is, the PZI results are more conservative for most crack sizes than those from **Eqs. (H24) (H25), and (H26)**, recommended by MPC **(Anderson, 1993)** and other documents [2].

In MPC **(Anderson, 1993)**, these formulas are assumed to be applicable to both cracks and grooves, and are claimed to be conservative. Such an approach, however, may not yield conservative results in the presence of cracks, as observed from the comparison. Consideration of the possibility of plastic zone instability may be necessary.

As discussed earlier, the r_Y at PZI, $(r_Y)_{\tan}$, and, correspondingly, $(\alpha/s^2)_{\tan}$, are uniquely determined for a given crack geometry. That is, the effect of α on σ/σ_Y at PZI, $s_{\tan}$, is proportional to $\sqrt{\alpha}$; $s_{\tan} \propto \sqrt{\alpha}$. In other words, the plane stress situations $(\alpha = 2)$ yield the most conservative results. For the thin-walled shell analyzed here, the plane stress conditions are expected to prevail except perhaps for very short cracks. As long as the same flow stress is used for σ_Y in both analyses, the results in **Table H.1** are good for direct comparison in plane stress conditions. Also note that the values of α have the opposite effect on the plastic zone instability and the fracture toughness.

The plastic zone instability analysis is based on r_Y-corrected effective linear-elastic fracture mechanics (and the material crack extension resistance curve). Therefore, PZI analysis requires only LEFM solutions. Accurate LEFM solutions are readily available for many crack configurations, in handbooks, such as this handbook, or are otherwise easily obtained.

The methods for determining the progressive increases of r_Y leading to the conditions for PZI are described in sufficient detail so that these procedures can be easily followed in the analysis of other crack problems.

Appendix I

Approximations and Engineering Estimates of Stress Intensity Factors

The uses of stress intensity factor values in engineering applications of fracture mechanics do not require values computed with absolute precision. Often estimates within 5% or better are sufficient for the purpose, as loads and material properties are seldom known with great accuracy. Therefore, the techniques of developing approximations for stress intensity factors for applications are discussed here. The solutions and formulas found earlier in this handbook for idealized crack and body configurations are explored for estimating stress intensity factors for the nonideal configurations often confronted in applications. Intuitive methods for estimating the magnitude of errors in such estimates will be emphasized. In addition, methods of developing bounds on values will be discussed. Indeed, the art of developing useful stress intensity factor estimates with the intuitive tools of linear elasticity and "strength of materials" approximations can be developed with practice for many advantages in applications.

DIMENSIONAL ANALYSIS AND OTHER ASPECTS OF STRESS INTENSITY FACTORS

The dimensional nature of stress intensity factors, K, is defined from their definition as expressed in the equation in Part I of this handbook. All formulas for K have units of force over length to the three-halves power. Therefore, if a remote stress, σ, is the applied load for a member containing a crack, then the K-formula for that situation will have the form

$$K = \sigma\sqrt{\pi a}\, f\left(\frac{a}{W}, \frac{B}{W}, \frac{L}{W}, \textit{etc.}\right) \qquad \textbf{(I1)}$$

whereas if a force per unit thickness, P, is the applied load, the form is

$$K = \frac{P}{\sqrt{\pi a}}\, g\left(\frac{a}{W}, \frac{B}{W}, \frac{L}{W}, \textit{etc.}\right) \qquad \textbf{(I2)}$$

where for simplicity only Mode I K is considered in both of these formulas and a, B, L, W, etc. are the crack size and other characteristic dimensions of each configuration considered. A quick review of a few solution pages will reveal these formats and alternatives for other types of applied loading, such as moment per unit thickness, M. The nondimensional "configuration functions," f ( ) and g ( ), also have special characteristic behavior depending on the type of configuration being considered. In particular, the finiteness of the body containing the crack, as described by the characteristic dimensions, often affects these formulas through separate factors, giving the forms

$$f\left(\frac{a}{W}, \frac{B}{W}, \frac{L}{W}, \text{ etc.}\right) = f_1\left(\frac{a}{W}\right) \cdot f_2\left(\frac{B}{W}\right) \cdot f_3\left(\frac{L}{W}\right) \cdot \text{ etc.} \tag{I3}$$

and

$$g\left(\frac{a}{W}, \frac{B}{W}, \frac{L}{W}, \text{ etc.}\right) = g_1\left(\frac{a}{W}\right) \cdot g_2\left(\frac{B}{W}\right) \cdot g_3\left(\frac{L}{W}\right) \cdot \text{ etc.} \tag{I4}$$

Reviewing a few solution pages also shows this to be characteristic of most of the K formulas. It is the intent here to exploit this tendency to make estimates of stress intensity factors.

EDGE CRACK

We begin by considering the configurations on solution **pages 5.1 and 8.1**. The discussion is restricted to normal stress, σ, on the boundaries. Note that the configuration of **page 8.1** can be formed from that of **page 5.1** by cutting the latter along its vertical axis of symmetry. However, before cutting, uniform stress, σ, on the boundary must be removed so that no (net) horizontal force is present on the free edge of configuration 8.1. This removal is illustrated by superposition in **Fig. I.1**, which shows that although no net force remains, a distribution of self-equilibrating normal stress is left on the vertical cut surface. The maximum intensity of this

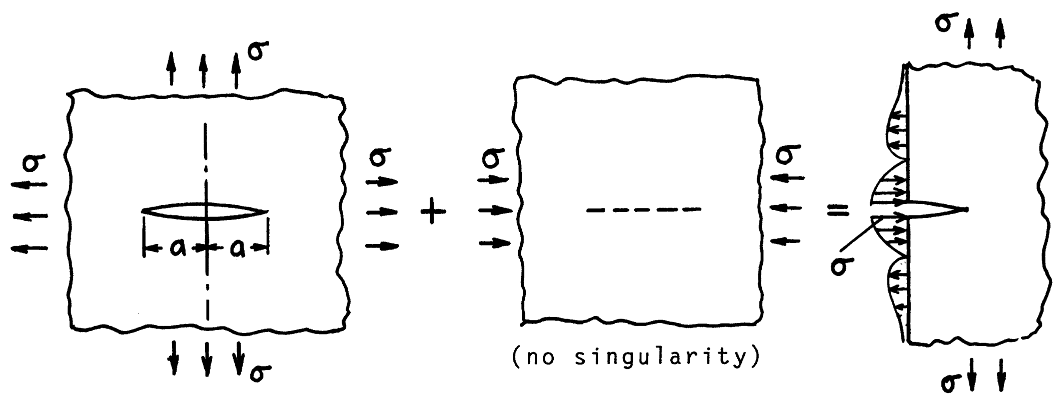


Fig. I.1.

stress is compression, - σ, near the emanation of the crack from the edge and it dies away to zero at locations far from the crack as compared with the crack size, a. These remaining stresses on the cut in **Fig. I.1** must also be removed to make it into the free edge of configuration 8.1. In order to remove these stresses on the edge, tension is required near the crack site and compression farther away, to net zero everywhere on the cut. Note that these added stresses required to make the cut a free surface would tend to open the crack more than before cutting. This implies that the K values should be higher for configuration 8.1 than for configuration 5.1.

Indeed, the factor is 1.1215 from the formula on solution **page 8.1**, or the difference in values is about a 12% increase in K due to the free-surface effect. Also notice that the introduction of the free surface caused a factor, f, to be incorporated in the K formula of the form

$$\begin{aligned} K &= \sigma\sqrt{\pi a}\cdot f \\ f &= 1.1215 \end{aligned} \qquad (\mathbf{I5})$$

Although this result was known from the solution pages, it illustrates the idea of using the 5.1 solution with a modifying factor to approximate solution 8.1. Note that a stress distribution on an edge perpendicular to the crack of maximum intensity equal to the applied stress caused a error of only 12%.

Next, consider the center cracked strip configuration on solution **page 2.1.** Again the solution for K is given, but it is informative to consider how it might be approximated if the solution was not known. On solution **page 7.1**, the closed-form solution is given for the repeated crack problem, which could be cut on successive axes of symmetry between cracks to form a strip. The stresses along the cuts can be computed from the stress function, Z, after again removing the uniform lateral stress, σ, as in the previous example. The final remaining self-equilibrating stresses on the cuts are again maximum at the crack line and of intensity given by

$$\sigma_x\left(\frac{W}{2},0\right) = \sigma\left[\frac{1}{\cos\frac{\pi a}{W}} - 1\right] \qquad (\mathbf{I6})$$

Thus as the crack size a goes to zero compared with the strip width, W, the stresses to be removed tend to zero. Moreover, for a crack length of one-half of the strip width, the maximum stress to be removed is $\sigma_x = 0.414\sigma$. Again note that removing these stresses would tend to open the crack further. Noting the magnitude of stresses to be removed compared to the edge crack example, up to a crack size of half the strip width the estimated error in K would be 12% times 0.414 or about a 5% underestimate. Therefore, historically, the K formula on solution **page 7.1** was considered a good approximation for the center cracked strip test configuration until a better formula became available. Indeed, we can improve it here by writing it as

$$K = \sigma\sqrt{\pi a}\cdot f_1\cdot f_2 \qquad (\mathbf{I7})$$

where

$$f_1 = \sqrt{\frac{W}{\pi a}\tan\frac{\pi a}{W}} \qquad (\mathbf{I8})$$

and knowing that f_2 should be asymptotic to 1 for a diminishing crack, then we estimate

$$f_2 = 1 + 0.05\left(\frac{4a}{W}\right)^2 \quad (\text{compensates for } 5\% \text{ error for } 4a = \mathrm{W}) \qquad (\mathbf{I9})$$

as an approximation. Now this result may not be as good as some of the better formulas given on solution **page 2.1**, but we can see that it is probably accurate within 1% for crack sizes up to more than half the strip width. Hence, without the benefit of other solution techniques, a very good "engineering estimate" is illustrated here. The purpose is simply to develop the method of forming good estimates.

In addition, this example also illustrates the method of using factors for various configuration effects. The factor, f_1, is the exact factor for the nearness of periodic cracks in an infinite sheet, and f_2 is the effect of cutting on the axes of symmetry between the cracks.

DOUBLE EDGE CRACKED STRIP

This configuration is also treated in Part II and presents an interesting example for estimation purposes. Notice that the configuration on solution **page 5.1** can be cut on the vertical axes of symmetry centering on two adjacent cracks to form the configuration. Again removing the uniform horizontal stress field prior to cutting, the remaining self-equilibrating normal stresses on the cut surfaces are a maximum of - σ compression at the locations of the cracks. For small cracks compared to the strip width, it is thus appropriate to use the 1.1215 factor for edge cracks. However, as the cracks deepen and the tips approach each other, the force transmitted from top to bottom of the strip is transmitted through a narrowing neck, which is already adequately represented in the solution without cutting. Therefore, the 1.1215 factor should diminish to 1 asymptotically as the crack tips approach each other. Consequently, an estimate of the appropriate effects is

$$K = \sigma\sqrt{\pi a} \cdot f_1 \cdot f_2 \tag{I10}$$

where

$$f_1 = \sqrt{\frac{2b}{\pi a}\tan\frac{\pi a}{2b}} \quad (2b \text{ is the strip width } W) \tag{I11}$$

and

$$f_2 = 1 + 0.1215\cos^n\frac{\pi a}{2b} \tag{I12}$$

where n is guessed to be 2 or greater. In Part II, Tada found, taking n = 4, results in less than 0.5% error over the full range of a/b. However, if we simply take the straight line approximation given by

$$f_2 = 1 + 0.1215\left(1 - \frac{a}{b}\right) \tag{I13}$$

the results are within 3%. Consequently, this example provides good approximations (even where the asymptotic behavior has been ignored). In the text of Part II for this configuration, other asymptotic approximations are given as examples of the possibilities.

As a further note, for deep cracks in the double edge cracked strip, that is, a → b, the above solution can be shown to agree with the closed-form solution of **page 4.9**. This fact adds to confidence through agreement with an exact limiting case. When developing approximate formulas, checking with limiting cases should be done whenever possible.

WEDGE FORCE SOLUTIONS

Although configurations with concentrated forces applied to the surface of a crack are seldom of direct practical interest, they are useful through superposition techniques in forming the solutions to many other configurations and in developing estimates of K. The technique involves first solving for the stresses present on the crack surface with the crack absent and then using the wedge force solutions to apply distributed forces (or effectively opposite stresses) to wipe out these stresses on the crack surfaces. This technique is shown schematically in **Fig. I.2**. The geometric configuration must be the same as the original configuration for the companion wedge force solution for superposition to apply exactly. However, for estimates, the companion wedge force configuration may differ slightly and with good judgment provide suitable approximations. This judgment can be developed through an understanding of "St. Venant's Principle" and like methods of intuitive stress analysis.

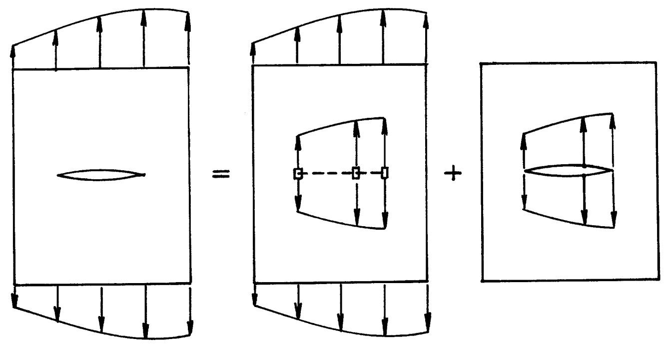


Fig. I.2

WEDGE FORCES NEAR THE TIP OF A CRACK

The wedge force solution for a semi-infinite crack with forces at distance, b, from the tip is given on page 3.6 and is

$$K = \frac{P}{\sqrt{\pi b}} \tag{I14}$$

where P is force per unit thickness. It is especially noted that forces nearer the crack tip are of greater influence due to the inverse $\sqrt{b}$ in the formula. Taking limiting cases of all other wedge force solutions will show this same influence as b becomes small compared with other dimensions. Therefore the stresses on the crack surface near the crack tip with the crack absent are of greatest influence. This fact is very useful in making estimates.

GREEN'S FUNCTIONS FOR STRESS INTENSITY FACTORS FROM WEDGE FORCE SOLUTIONS

For internally cracked infinite sheets, the method illustrated by **Fig. I.2** can be used with the wedge force solutions on **pages 5.10 and 5.11** to write forms that can be integrated to obtain K. In each of these solutions, the load P may be replaced by $\sigma_y dx$, and b with x for the integration, where $\sigma_y = \sigma_y(x, 0)$ is the normal stress on the crack surface with the crack absent. For stress distributions, σ_y, which are not symmetric with respect to the y-axis, from **page 5.10**:

$$K = \frac{1}{\sqrt{\pi a}} \int_{-a}^{a} \sigma_y \sqrt{\frac{a+x}{a-x}}\, dx \tag{I15}$$

For stress distributions symmetric with respect to the y-axis, from **page 5.11**:

$$K = 2\sqrt{\frac{a}{\pi}} \int_{0}^{a} \frac{\sigma_y}{\sqrt{a^2 - x^2}}\, dx \tag{I16}$$

Note that for uniform stress, σ, applied to the sheet, the crack absent stress on the crack surface is $\sigma_y = \sigma$ (constant), which may then be moved outside the integral sign, whereupon integration gives $\frac{\pi}{2}$. The net result is the familiar form

$$K = \sigma\sqrt{\pi a} \tag{I17}$$

Although this result is trivial it verifies the method. More important is to note that for any crack absent stress distributions, σ_y, the integrals may be evaluated numerically, if not analytically. This is the reason that many wedge force solutions are given in this handbook for other configurations. They will be used later in this discussion on estimating.

Wedge force solutions are also given for many three-dimensional problems (see solution **pages 23.1, 24.3, 24.24, 24.25, 25.1**, etc.). In many cases, other solutions have been generated from these solutions (see adjacent solution pages) using the Green's function method for the same geometrical configuration with different loadings. These solutions are of assistance in estimating as well.

PRELIMINARIES FOR ESTIMATING SOLUTIONS FOR THREE-DIMENSIONAL CRACKS

The plane problems presented here are stress solutions that are independent of the thickness of the sheet. Consequently, they are really three-dimensional solutions which simply do not vary in the out-of-plane direction. However, for truly nonvarying conditions in the out-of-plane direction, the "constraint" conditions must be either plane stress or plane strain. With sheet thicknesses of the order of the crack size, a (or other significant planar dimensions), neither plane stress nor plane strain fully applies. Indeed, in regions of high planar stress gradients, the inner sheet conditions may tend toward plane strain with plane stress near the surfaces. The crack tip is just such a high-stress gradient region. Therefore, the K formulas have inherent error due to these constraint effects. However, if we view the in-plane displacements as being compatible (approximately equal through the sheet), then the stress differences are at most by the factor $1 - \nu^2$, where ν is Poisson's ratio. Therefore, for a perfectly straight crack front through a sheet of finite thickness, at some portions of that crack front our K formulas may be incorrect by a maximum equal to that factor (i.e., less than 10% error maximum) and for the largest errors only a very small portion of the crack front would be involved. Therefore, the application of our "exact solutions" for K inherently involve some error. Moreover, crack fronts though the thickness of a sheet are never perfectly straight. Nevertheless it is the final objective to make "estimates" of reasonable engineering precision and feasible accuracy.

INTERIOR "PENNY-SHAPED" CRACK

The exact solution for the penny-shaped (circular), crack is found on solution **page 24.1**. Suppose that the solution for K is not known and we wish to estimate it for practice in forming estimates. Referring to **Fig. I.3**, the K formula must contain a factor proportional to the applied stress, σ, and also to the square root of the only characteristic dimension, a. Then the only remaining element is to find the proper numerical factor, f, in the following form

$$K = \sigma\sqrt{\pi a} \cdot f \tag{I18}$$

Now if the crack were a straight through-the-body "tunnel crack" of width $2a$, then f would be 1. Viewing the circular crack as made from the tunnel crack but with portions of the crack surface pulled closed, we can see that f must be less than 1 for the circular crack. In considering the technique of wiping out the stresses on the crack surface with the crack absent described in the previous paragraphs, we note that uniform stresses would be present. For the ratio of the area of crack surface to the length of crack front (about which the load transmission interrupted by the crack must be carried), note that the circular flaw has a ratio of one-half that of

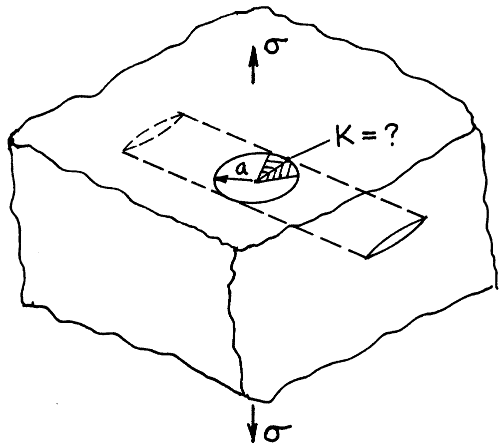


Fig. I.3

the tunnel crack. Based on that ratio, our first estimate of the appropriate f might be $1/2$. However, we note that the area of the circular crack is more closely gathered about its crack front. Our discussion of wedge forces noted that interruption of forces closer to the crack front has greater influence on K. Therefore our estimate of f should obviously be more than $1/2$; so a first guess might be about $2/3$. The exact solution for f shows this intuitive estimate to be quite good. Comparing shows

$$f = \frac{2}{\pi} = 0.6366 \cong \frac{2}{3} \quad (\text{within } 5\% \text{ error}) \tag{I19}$$

With a little practice in guessing we would conclude that 0.6 would be too small and $3/4$ would be too high and conclude that the guess of $2/3$ is a reasonable estimate between these "bounds," even if the exact solution is unknown. (If you agree, you are ready for further developments of techniques of estimating.)

INTERNAL ELLIPTICAL CRACK (ESTIMATED SOLUTION)

Consider a flat elliptical internal crack subject to uniform tension normal to the crack. Let the smaller of the semi-axes of the ellipse be described by a and the greater semi-axis by b. If $a = b$ it would be the circular crack previously discussed, and $b \to \infty$, then it becomes the tunnel crack. The K along the crack front of the ellipse will vary but is obviously a maximum at the semi-minor-axis location since more of the area of crack surface is nearby per increment of crack front length. Suppose for practical reasons the maximum K is desired. The previous formula for the circular crack is appropriate with f assumed to be a nondimensional function of the ratio a/b. A first estimate might be to assume that f varies approximately linearly from 1 for $a/b = 0$ to $2/\pi$ for $a/b = 1$. This can be expressed by

$$f = 1 - \left(1 - \frac{2}{\pi}\right)\frac{a}{b} = 1 - 0.363\frac{a}{b} \tag{I20}$$

Now for $a/b = 1/2$ this estimate gives $f = 0.818$, whereas the exact solution gives $f = 0.826$, which indicates about a 1% error (not only a good guess but a very lucky guess!). Perhaps over the full range, it can be simply said that the error is expected to be within 5%.

INTERNAL ELLIPTICAL CRACK (EXACT SOLUTION)

The full, exact solution for the internal elliptical crack subject to uniform remotely applied stresses is given on solution **pages 26.1 through 26.4**. The variation of K for all modes is given for points on the crack front around the ellipse. For practicality, the discussion here is restricted to applied normal stress perpendicular to the crack, which results in first mode K only. For this configuration, the result from **page 26.2** with a and b interchanged is

$$K = \frac{\sigma\sqrt{\pi a}}{E(k)}\left(\sin^2\theta + \frac{a^2}{b^2}\cos^2\theta\right)^{1/4} \qquad (\mathbf{I21})$$

where $E(k)$ is a complete elliptic integral of the second kind depending only on the a/b ratio, which is found in **Appendix L**, and θ is the parametric angle for the ellipse as on **page 26.1**. Of special interest in estimating K for imperfect elliptical shaped cracks is l, which is the length of the normal (from the ellipse itself to an intersection with its major axes). This length is also illustrated on **page 26.1**. As noted, it can be expressed by

$$l = a\left(\sin^2\theta + \frac{a^2}{b^2}\cos^2\theta\right)^{1/2} \qquad (\mathbf{I22})$$

The expression for K then simplifies to give

$$K = \frac{\sigma\sqrt{\pi l}}{E(k)} \qquad (\mathbf{I23})$$

This formula demonstrates that K values for irregular internal flaws are dominated by the local geometry near the crack front region being considered through the strong dependence on l. On the other hand, the dependence on the overall crack shape through $E(k)$ is quite limited since its values only vary from 1 to 1.57 over extreme changes in shape. This concept is useful in making K estimates for both irregular internal cracks and surface cracks of various kinds.

SEMI-ELLIPTICAL SURFACE CRACK

Figure I.4 shows a semi-elliptical surface crack perpendicular to a flat surface with remotely applied normal stress parallel to the surface and normal to the crack. It is chosen as a configuration that is typical of many flaws found in practice that are critical in impairing the strength of structures. These flaws are almost always two or more times as long on the surface from which they emanate as they are deep. Although no completely closed-form solution determined analytically is available for this type of surface crack, it has been discussed extensively in the literature by complicated semianalytical techniques and numerical methods, which almost always have hidden assumptions not accessible to the user. For this reason semi-elliptical surface cracks are good candidates for checking by K estimating procedures that are convenient, quick, and have no hidden assumptions. Therefore this problem is discussed as a first example of estimating for surface cracks.

The standard approach here is to apply the usual form of

$$K = \sigma\sqrt{\pi a}\cdot f_1 \cdot f_2 \qquad (\mathbf{I24})$$

where the geometrical factors, f_1 and f_2, are the effect of the elliptical shape, a/b, and the effect of introducing the front free surface, respectively. The first factor is taken from the exact elliptic solution as

$$f_1 = \frac{1}{E(k)} \qquad \text{(exact)} \qquad (\mathbf{I25})$$

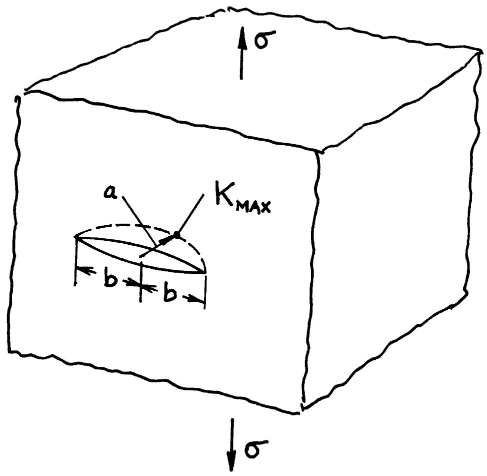


Fig. I.4

The second factor is that the free surface, which tends to 1.1215 for long surface cracks, that is, $a/b \to 0$, and would tend toward 1 for semicircular shapes, $a/b \to 1$, considering that the tendency for a surface crack to open further than an internal crack would be almost entirely suppressed for the semicircular crack. Therefore, it seems reasonable to estimate the second factor as

$$f_2 = 1 + 0.1215\left(1 - \frac{a}{b}\right) \quad \text{(estimate)} \tag{I26}$$

This estimate is surely within 5% error and probably much better than that if good intuition prevails. Furthermore, there have been no hidden errors, numerical or otherwise, in our estimate; this fact instills confidence.

SEMI-ELLIPTICAL SURFACE CRACK IN A FINITE THICKNESS PLATE

The ASME Nuclear Pressure Vessel Code requires that postulated semi-elliptical surface cracks in the wall of a vessel, of a depth of $1/4$ of the wall thickness and of an aspect ratio a/b, are to be analyzed. Therefore our interest in semi-elliptic surface cracks in finite thickness plates is evident. **Figure I.5** shows the configuration, including the finite thickness, t, the effect of which is another factor $f_3(a/t)$, beyond those in the previous analysis of semi-elliptic surface cracks. Indeed, f_2 and f_3 should be considered together in the analysis, since their effects are physically coupled. As an alternative, writing separate factors, their combined error effects should be considered together to form a total error for the separated functions. Now for a surface flaw in a finite thickness plate, f_2 may be noted to be too large a correction by perhaps 5% at most. On the other hand, for the f_3 correction factor for a crack tip approaching a back free surface, the “square root of tangent” correction previously applied to the central crack strip estimate was seen by itself to undercorrect by about 5%. Therefore, by using these two corrections together in this analysis, we create compensating errors well within 5%. For this reason, we choose f_3 as (f_1 in the strip example)

$$f_3 = \sqrt{\frac{2t}{\pi a}\tan\frac{\pi a}{2t}} \tag{I27}$$

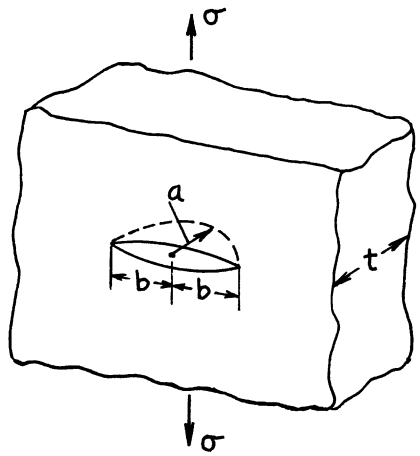


Fig. I.5

This combined with the immediately preceding analysis leads to

$$K = \frac{\sigma\sqrt{\pi a}}{E(k)}\left[1 + 0.12\left(1 - \frac{a}{b}\right)\right]\sqrt{\frac{2t}{\pi a}\tan\frac{\pi a}{2t}} \tag{I28}$$

(estimated error $< 5\%$ for $a/t < 1/2$)

This estimating formula for semi-elliptic surface cracks in finite thickness plates was first proposed by **Paris (1965)**. **Newman (1979b)** used finite element analysis to compare Paris' formula with about 10 alternative formulas. He found Paris' formula to be the second most accurate of all of the formulas, many of which had complicated mathematical justifications associated with them. The simplicity of the logic and the freedom from hidden assumptions for the this formula makes it even more appealing.

IRREGULARITIES IN SHAPE OF ALMOST SEMI-ELLIPTICAL SURFACE CRACKS

Real surface cracks invariably are not of a perfect semi-elliptical shape, but almost all analyses of them assume they are perfect without special justification or consideration. It is most noticeable that the ends of these cracks, where they meet the surface, are almost never perpendicular to that surface. At other places along such a crack front the curvature also rarely perfectly matches the ellipse chosen. To better evaluate K along crack fronts of such irregularly shaped cracks it is desirable to discuss some possibilities for improvements.

A first example is based on the actual shape of a preexisting crack in an $F - 111$ wing box that led to the loss of this aircraft in late 1969 and the grounding of this type of aircraft for more than 6 months. The crack is carefully drawn in **Fig. I.6** to illustrate its actual shape. In addition, the figure shows two perfect ellipses, one matching the length and depth of the real flaw, and the other matching the depth and curvature of the real flaw at the deepest point where K is anticipated to be maximum. The reason for selecting these two ellipses is that the first will give an "upper bound" on the K value and the latter will give a "lower bound" of the value and a better estimate as well. The second ellipse is a closer estimate because it better approximates the local geometric conditions at the point where K is desired.

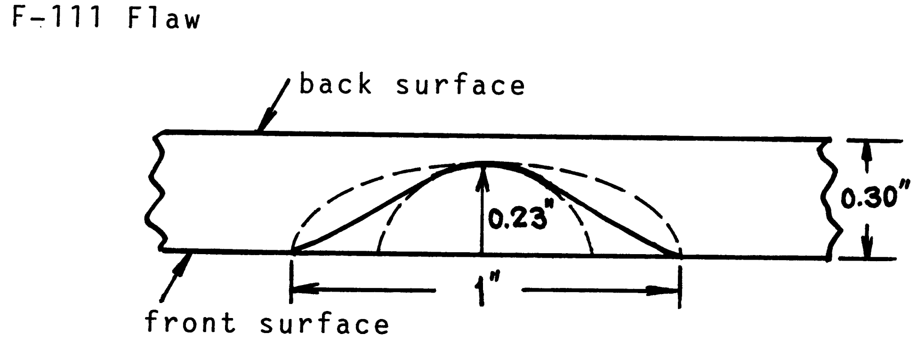


Fig. I.6

Taking the approximate dimensions from the ellipses and computing the correction f for both results in

$$K = (1.18 \textit{ to } 1.34)\sigma\sqrt{\pi a} \tag{I29}$$

This shows a difference of about 15% in the coefficient f. The correct result is undoubtedly closer to the lower value but also surely above it (rather than the upper value, which most often is blindly adopted). Thus if we elect to estimate the proper value at 1.24 it would be 5% above the lower bound and 9% below the upper bound, and likely within $\pm$ 3% or probably better for the geometrical correction factor f. Now the basis of the formula was no better than $\pm$ 5% in the first place, so the maximum combined error would be $\pm$ 8%. Knowing these bounds, the 1.24 factor could be intelligently adjusted up or down for a particular application depending on whether a low or high estimate is dictated by that application. Although engineering practice often simply computes a single "answer," the bounding technique and error estimates provided here have obvious advantages. Furthermore, in a real analysis of strength impairment of a structural component, the precision of the values of applied stresses and material properties should also be considered in a consistent fashion. Such a practice was actually followed in assessing the $F - 111$ failure.

ESTIMATES OF K NEAR THE INTERSECTION OF A SURFACE CRACK WITH THE SURFACE

Two difficulties in estimating K occur near the intersection of a surface crack with a surface. First, the crack tip stress field and its constraint parallel to the crack front change from plane strain in the interior to plane stress at the surface. Moreover, plastic flow near a crack tip will modify this constraint so that detailed solutions by purely elastic analysis for constraint effects will themselves be inaccurate. However, the difference between plane stress and strain is obviously a maximum factor in stress or K results approaching $1 - \nu^2$. That is to say, at the plane stress region near the surface, K formulas may give up to about a 9% overestimate of K. Notice that near surfaces, fatigue cracks frequently lag behind interior growth, which is explained at least in part by this effect. The second difficulty is simply that flaws almost never intersect the surface with their fronts perpendicular to the surface. Even when almost semi-elliptic surface flaws intersect a surface, near that intersection, the curvature of the crack front does not match that of a perfect semi-ellipse. This difficulty in assessing the local K can produce large errors if the formulas for perfect ellipses are blindly applied. This second difficulty can be remedied by suitable estimating procedures for K.

It is noted from solution **pages 26.1 and 26.2** that the formula for K near the narrow end of an ellipse is

$$K = \frac{\sigma\sqrt{\pi\rho}}{E(k)} \tag{I30}$$

where ρ is the end radius of the ellipse. For almost semi-elliptic surface cracks where they do meet the surface in a perpendicular manner this formula shows that K is strongly dependent on the local crack front radius ρ and more weakly dependent on the gross proportions of the ellipse through $E(k)$, since it only varies from 1 to 1.57 for extreme changes in proportions from infinitely long to circular. Consequently, the above formula for K should be used with the local radius of the actual elliptical shape. This will normally give reasonable results when the crack front varies somewhat from perpendicular to the surface.

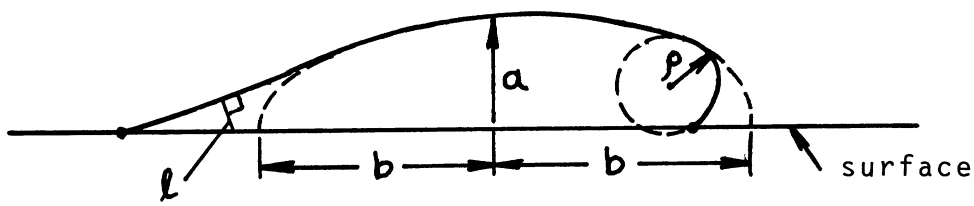


Fig. I.7

For more extreme cases of lack of perpendicularity of the crack front to the surface, further considerations are appropriate. **Figure I.7** illustrates cases of interest. The left and right intersections with the surface (solid curve) are the extreme cases. The theoretical elastic stress intensity values, K, for the intersection points are actually zero for the left nonperpendicular type of intersection and infinite for the right one. However, practical treatment and judgment of the relevant values would probably dictate using the local radius ρ in the above formula for the right crack end, along with an $E(k)$ using the proportion of the dashed ellipse. On the other hand for the left crack end, it would be appropriate to substitute l, the length of the perpendicular from a local portion of the crack front, for ρ in the above formula. Solution **pages 26.1, 26.2, and 26.6** support this suggestion. Of course, corrections for front free surface and back free surface, as with previous estimates, should be added if finite thickness plate is involved. However, the back free surface correction would be less influential at the crack ends, so l or ρ should be used in the tangent correction rather than a.

It needs to be emphasized here that, in most treatments of irregular semi-elliptical surface cracks, and in many computational programs, the effects of the irregularities discussed here have been simply ignored. As noted, these effects can be significant even for moderately irregular cracks.

GROSSLY IRREGULAR SURFACE CRACKS

For grossly irregular surface cracks, it is reasonable to use the intuitive knowledge gained from the preceding discussion to make estimates of stress intensity, K, at points along the crack front. **Figure I.8** is an example of such an irregular surface crack. For the initial discussion of this example, it is assumed that uniform normal stress, σ, normal to the crack is applied remotely to this region of the body. The objective here is to estimate K on each segment of the crack front between points labeled A through H. With each segment, the logic of the estimate is explained to assist the reader in understanding estimating.

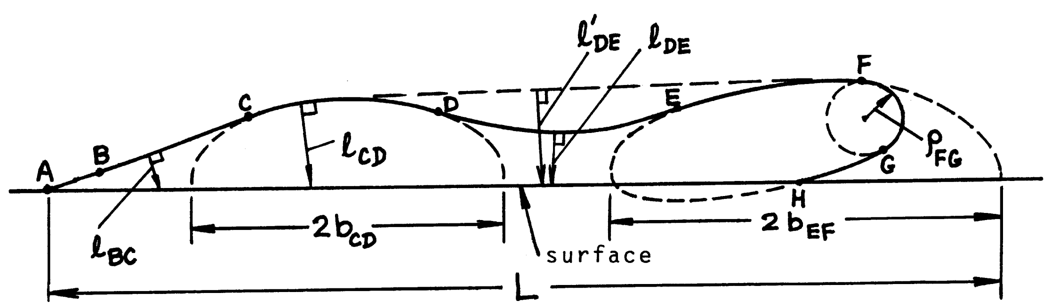


Fig. I.8

$A - B$: Along this segment, the K tends to zero at A and the relatively large crack to the right may be regarded as infinitely large. The K is then estimated as if it were an infinitely long surface (edge) crack as

$$K_{AB} = 1.1215\sigma\sqrt{\pi l} \tag{I31}$$

where the crack depth l is most appropriately taken as locally normal to the crack front for the point of interest along $A - B$.

$B - C$: By the time it reaches point C, the crack is deep enough that the total length, L, is finite by relative proportions so that a correction for finite length such as $E(k)$ is appropriate. Since $E(k)$ itself only varies between 1 and 1.57 for extreme changes in proportions, a/b, the estimation of the proportion to use is not critical. As the proportions are varied from fitting the curve of segment $C - D$ to using the full length of the surface crack, L, the $E(k)$ varies from about 1.21 to 1.07. Noting that the effective length is somewhere midway between these extremes, selecting approximately 1.14 is surely within 3% of correct. The 1.1215 surface correction is also reduced by the finite length toward a minimum of 1 for a semi-circular crack. Obviously, this correction should be quite close to 1.1215, so we choose 1.09 as surely within 2% of correct. The overall correction is thus

$$\frac{1.09}{E(k)} = \frac{1.09}{1.14} = 0.96 \tag{I32}$$

within a maximum error of 5% (really believed to be within 3%). Therefore as the point of interest moves from B to C, the K formula would appropriately change gradually (respectively) as

$$K_{BC} = (1.1215 \text{ } to \text{ } 0.96)\sigma\sqrt{\pi l_{BC}} \tag{I33}$$

This should allow estimating K on this segment within 5%.

$C - D$: For this segment, the K values would be best approximated by the dashed ellipse fitting the segment, with the following modifications. The front free surface correction should be reduced to the 1.09 value previously elected, and the $E(k)$ value should be modestly reduced from the 1.21 value for this ellipse. Electing a reduction to about 1.15 is surely within 3% of correct. The result is

$$K_{CD} = 0.95\sigma\sqrt{\pi l_{CD}} \tag{I34}$$

For this segment the values of K are surely well within 4% of correct with the worst error likely approaching D. Since the values that will result from this formula are considerably larger than those from the formulas for earlier segments due to larger values of l, its precision would probably be more important in a real application. Indeed, for this configuration of crack, the maximum K probably occurs on this segment and precision is therefore most important here.

$D-E$: Estimating K over this segment poses special problems due to the reversed curvature of the crack front. In particular the surface correction would gradually exceed the 1.1215 value with its maximum at the nearest point of the crack front to the surface, because the crack is deeper on both sides of this point. Accounting for the finite length, L, for this configuration would reduce the 1.1215 factor to a minimum of about 1.09 using previous forms. Therefore, at the portion of this segment nearest the surface it can be stated that

$$K_{DE}(\min) \geq 1.09\sigma\sqrt{\pi l_{DE(\min)}} \tag{I35}$$

On the other hand, taking the dashed straight line joining the peaks in the crack depth and using l' at that same location would indicate a maximum K, again adjusting for finite length, as

$$K_{DE}(\max) \leq 1.03\sigma\sqrt{\pi l'_{DE}} \tag{I36}$$

At this location l' is noted to be about twice the size of l. Accounting for the numerical coefficients in these expressions, the second exceeds the first by about 33%. Intuitively, the correct result should be closer to the minimum than the maximum, since it clearly is a closer representation of the real configuration. Hence, the estimate should increase the minimum by considerably less then half of 33%, so 10% is taken. Based on this logic, the K at the nearest point on $D-E$ to the surface is estimated as

$$K_{DE} = 1.20\sigma\sqrt{\pi l_{DE(\min)}} \quad \text{(nearest point)} \tag{I37}$$

This estimate is also believed to be within 5% of the correct result.

As a check on this result the notched round bar solution on **page 27.1** may be consulted. It has a crack front with reversed curvature as is the case for the nearest location on segment $D-E$. Matching this crack front curvature in a notched round bar and taking $l_{DE(\min)}$ as the crack depth in the notched round bar (estimated as $a/b = 0.75$ for the round bar) results in a coefficient of 1.21 instead of the 1.20 estimated above. This degree of agreement is somewhat fortuitous, since the notched round bar does not have increasing crack depth as the point is moved away from the nearest to surface location and neither does it have a finite length crack, which would cause errors to be somewhat compensating. However, the agreement is acceptable and helpful in giving confidence in the claim of 5% accuracy.

$E-F$: For this segment the suggested procedure is to consider both the dashed ellipse best fitting points $E-F-G-H$, with emphasis on fitting the curvature of $E-F$, and also taking the dashed semi-ellipse fitting $E-F$. For the former, the free surface correction should acknowledge more effect at E diminishing toward F, from a value exceeding 1.1215 to 1, respectively. On the other hand, for the latter semi-ellipse, the results at E should be modestly reduced considering the gross deviation in shape at the right end and this reduction should be increased significantly approaching F. Indeed, upon reaching F the results of the first complete ellipse should be considered as the relevant estimate, whereas the estimates of the semi-ellipse are most relevant at E. Estimating the free surface correction for the semi-ellipse at E to be about 1.08 should give results within 5%. Estimating the full dashed ellipse's surface correction to be entirely absent or 1 should give results within 5% at F. It then remains to simply interpolate between the two for points E and F smoothly. This is left to the reader.

$F-G$: Between F and G the curve should be best fit with a radius, ρ_{FG}, which, using the full dashed ellipse, leads to

$$K_{FG} = \frac{\sigma\sqrt{\pi\rho_{FG}}}{E(k)} \tag{I38}$$

Because this segment is away from the influence of the free surface, it has not been corrected for it. However, since there is some remote free surface, it is a lower bound of the K estimate but certainly well within a 5% error of the actual value. A factor of 1.02 could be added to this expression, which would imply the accuracy then to be within 1% (if the inherent error is ever that small).

$G-H$: Before even getting to G along $F-G$, a correction for a crack approaching a back free surface might be considered. However, it was omitted here because conditions along $G-H$ become exceedingly difficult for simple estimating. First, it should be noted that K tends to infinity at reversed intersections with the surface, such as at H. In any analysis of a real problem, the ligament between the crack and the surface would be subject to stresses causing yielding to spread away from H, perhaps beyond G, depending on nominal stress levels. Therefore, it may be dangerously irrelevant to attempt an estimate. With that in mind, we proceed assuming that elastic action prevails. The opened part of the crack in the region between $E-F-G-H$ would shed load onto the uncracked ligament adjacent to $G-H$. Crudely, the stress interrupted by half of the width of the dashed full ellipse would be dumped through that ligament. The net section stresses there would thereby increase from approximately 3σ at G toward infinity at H. These increased ligament stresses could then be used in a solution, such as on **page 9.2**, to estimate K. Such a procedure is omitted here, as large errors would be present and the relevance is doubtful.

It might be more relevant to expect cracking of such a ligament in any application and anticipate the crack to grow in the end shape of the semi-ellipse shown. That is, we could start our estimates from some intermediate shape that would be easier to analyze with less loss of relevance. This also is left to the reader.

ADJUSTMENTS FOR STRESS GRADIENTS IN ESTIMATES OF K FOR SURFACE CRACKS

Generally with cracks absent, the stresses in bodies containing cracks vary gradually and continuously throughout the region where a crack appears. Exceptions are left for more advanced individual treatment. Restricting the discussion to normal stresses perpendicular to a crack, first consider the variation in stress, σ, along a surface in the direction of its intersection with the crack. **Figure I.9** shows the crack under consideration.

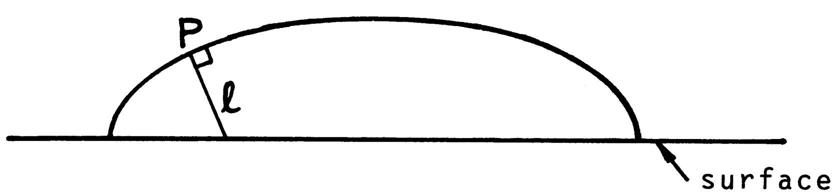


Fig. I.9

The analysis of the behavior of K along an elliptical crack shows a strong dependence on local stress conditions. For a point P, as on **Fig. I.9**, the stresses are on and near the normal to the crack front, l. This local stress dependence is also borne out by the character of K caused by wedge force solutions in both two and three dimensions as noted on solution **pages 3.6** and especially **page 23.1**, as well as others. Therefore, the simple rule to accommodate gradients of stress along the surface is simply to <u>take the average stress, σ, on l with the crack absent to evaluate K at point P.</u>

Furthermore, if stress gradients occur normal to the surface, additional consideration should be superimposed. **Fig. I.10** illustrates a linear gradient normal to the surface that is divided into the constant stress occurring at the crack front point, P, of interest and the additional stress away from that point. Now the K due to each of these distributions should simply be added. The K for the constant stress is as treated earlier. For the additional linearly varying stress normal to the surface with zero at the crack front, it is helpful to review the solutions on **pages 8.1 and 8.6, and 28.1 and 28.2** and to note the effect of stress gradients normal

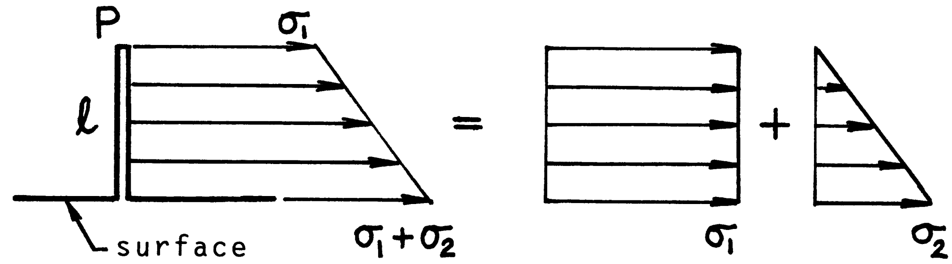


Fig. I.10

to the surface for both infinitely long surface cracks (edge cracks) and the other extreme of semi-elliptical cracks (i.e., semicircular cracks). The results are that an equivalent stress σ can be found for insertion into formulas for uniform stress, where for long cracks it gives

$$\sigma = \sigma_1 + 0.39\sigma_2 \tag{I39}$$

and for semicircular cracks it gives

$$\sigma = \sigma_1 + 0.30\sigma_2 \tag{I40}$$

Now these forms are so similar that it is relevant to suggest for elliptical cracks that a suitable approximation is

$$\sigma = \sigma_1 + \left[0.30 + 0.09\left(1 - \frac{a}{b}\right)\right]\sigma_2 \tag{I41}$$

This form is obviously quite accurate since it is exact at both extremes of elliptical shape and the adjustable term containing $^a/_b$ is very small for stress distributions with σ_2 of the same magnitude or smaller than σ_1. Thus any error from this factor would normally be within 2 or 3%. Moreover, using the stress distribution along l in **Fig. I.9** is entirely appropriate within the stated degree of accuracy. Finally, if the stress distribution with the crack absent varies in a nonlinear manner, as in **Fig. I.11**, then taking the extreme linear approximations using σ_2 or σ'_2 will give error extremes between which suitable judgments can be made with assured accuracy. This method accommodates having continuous stress gradients both along and into the surface. However, if the stress distributions are sharply peaked at the surface, such as near a notch, or otherwise not a smooth stress distribution, then estimating should be approached with extra caution. An example follows.

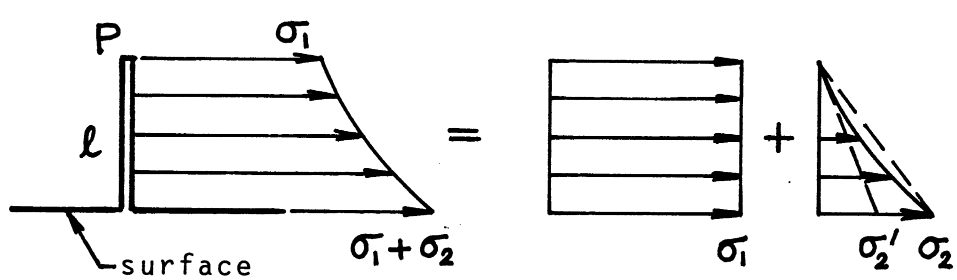


Fig. I.11

CRACKS EMANATING FROM A ROUND HOLE IN A PLATE IN TENSION

Figure. I.12 shows a circular hole in a plate subject to uniform uniaxial tension, σ, remotely applied. Assuming symmetrical through-thickness cracks exist at the maximum stress concentration points on opposite sides of the hole, it is desired to determine the K values for this case. Although a solution appears on **page 19.1** for this configuration, the estimating method as developed in the previous paragraphs here will be compared with this solution to evaluate the estimating method with nonlinear stress gradients.

With the crack absent, the stress distribution for normal stress on the crack plane is described by

$$\sigma_y(r) = \sigma\left[1 + \frac{1}{2}\left(\frac{R}{r}\right)^2 + \frac{3}{2}\left(\frac{R}{r}\right)^4\right] \tag{I42}$$

From this expression it is noted that a sharp peak in stress, of a value 3σ, exists next to the hole, which nonlinearly diminishes to a constant value, σ, away from the hole. The crack emanates from the surface of the hole that is not flat so that the 1.1215 surface factor should diminish rapidly away from the hole but in an unknown manner. Therefore, using that factor without diminution would cause an overestimate in K as the crack lengthens. Consequently, the previous method, using σ'_2, which would cause underestimates, is used with that factor for the evaluation. The resulting form is

$$K = 1.1215\left(\sigma_1 + 0.39\sigma'_2\right)\sqrt{\pi a} \quad \left({}^{a}/_{R} \ll 1\right) \tag{I43}$$

On the other hand, for larger cracks, the hole itself may be regarded as part of a longer crack of length $2a + 2R$ in a sheet with uniform tension. This can be evaluated with the familiar form

$$K = \sigma\sqrt{\pi(a+R)} = \sigma\sqrt{\pi a}\cdot\sqrt{1+\frac{R}{a}} \quad \left({}^{a}/_{R} \gg 0\right) \tag{I44}$$

Now both of these formulas for K will tend to give exact results, the first as the crack size approaches zero and the second for very large cracks (i.e., a large compared with R). In the intermediate region, both tend to give overestimates of the real K. Therefore, electing to take the smaller value of the two would tend to give the most reasonable results. The results are tabulated in **Table I.1** and are compared with Tada's best fit form from **page 19.1**.

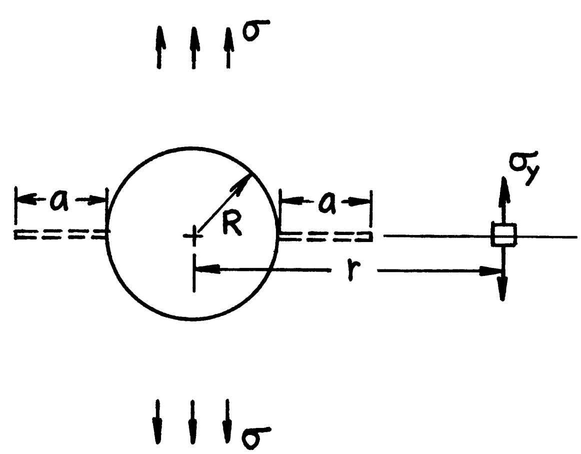


Fig. I.12

Table I.1

$\frac{a}{R}$	$1.1215\left[\frac{\sigma_1}{\sigma}+0.39\frac{\sigma'_2}{\sigma}\right]$	$\sqrt{1+\frac{R}{a}}$	Tada p. 19.1	% Error	Green's Function
0	3.3645	(infinite)	3.3645	0%	
0.02	3.2804		3.2357	+1.4%	
0.05	3.1422	(4.5826)	3.0614	+3.0%	
0.10	2.9494	(3.3166)	2.8131	+4.8%	
0.15	(2.7822)	2.7689	2.6079	+6.2%	(2.2397)
0.20	(2.6359)	2.4495	2.4357	+0.6%	(2.1305)
0.30		2.0817	2.1679	-4.1%	1.9554
0.50		1.7321	1.8246	-5.3%	1.7189
0.70		1.5584	1.621	-4.0%	1.5697
1.00		1.4142	1.444	-2.1%	1.4301
Infinite		1.0000	1.000	None	1.0000

These results speak for themselves for confirming the method of estimating in this difficult case. Other improvements could be attempted, but this seems sufficient to demonstrate the method. For example, the second integral **(I16)** can also be applied by regarding the hole as part of the crack. It will give even better values at intermediate crack sizes, that is, for $^a/_R$ of about 0.50 to 3.0 or more. Defining the apparent half-crack length as $l = a + R$, the appropriate form is

$$K = 2\sqrt{\frac{l}{\pi}}\int_R^l \frac{\sigma_y(r)dr}{\sqrt{l^2 - r^2}} \tag{I45}$$

where $\sigma_y(r)$ is given by **(I42)**.
Integrating, the numerical values are the final column in **Table I.1**.

Appendix J

Rice's *J*-Integral as an Analytical Tool in Stress Analysis

The J-Integral, which is most often associated with elastic–plastic fracture mechanics, is also a useful tool for linear and nonlinear elastic stress analysis of cracks. It is therefore discussed here, mostly in the latter elastic context, but with some comments on elastic–plastic analysis as well. It has also often been used as a powerful analytical tool in combination with finite element analysis, which shall get only minimal comment here.

The path-independent integral form of J, as illustrated by **Fig. J.1**, is given by **Rice (1968a)** as

$$J = \oint_{\Gamma} \left(W dy - T_i \frac{\partial u_i}{dx} ds \right) \tag{J1}$$

where

$$W = \int_0^{\varepsilon_{ij}} \sigma_{ij} d\varepsilon_{ij} = \text{ elastic strain energy per unit volume or work per unit volume if plastic}$$

$$T_i = \sigma_{ij} n_j = \text{ traction on the contour } \Gamma$$

and

$$n_j = \text{ components of the unit vector normal to contour}$$

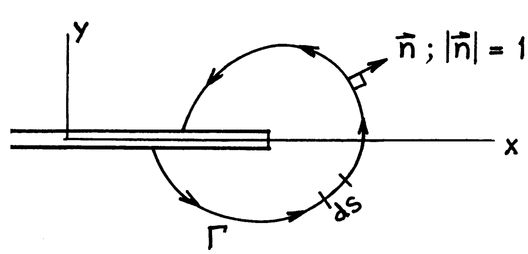


Fig. J.1

With the usual assumptions of equilibrium, small strains and rotations, and assuming that elastic behavior or deformation theory of plasticity applies, then

$$W = W(\varepsilon_{ij}) \quad \text{or} \quad \frac{\partial W}{\partial \varepsilon_{ij}} = \sigma_{ij} \tag{J2}$$

These assumptions, along with the Green-Gauss Theorem, allow the proof that the J-Integral is path independent when integrating on any contour, Γ, from the lower to upper crack surfaces, provided that the material is a continuum and that no singularities or loads are enclosed within Γ, except the crack tip itself.

PHYSICAL INTERPRETATION OF *J* FOR NONLINEAR ELASTIC CONDITIONS

The crack tip with the contour attached may be advanced by an increment, *da*. The terms in the J-Integral interpreted as a result of that advance are illustrated in **Fig. J.2**. **Rice (1968a)** notes that the value of

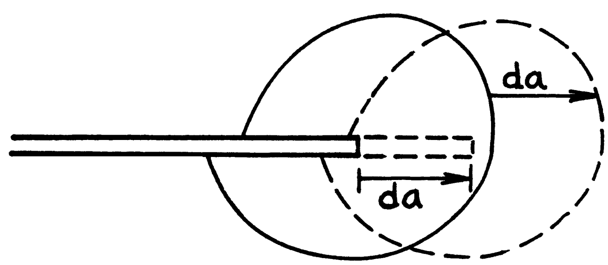


Fig. J.2

the J-Integral is the amount of energy pouring through the contour per unit increase in crack area, as characterized by *da.* This means that for nonlinear or linear elastic conditions, J gives the Griffith energy rate per unit increase in crack area, or

$$J = G = \frac{K^2}{E'} \text{ (elastic: linear or nonlinear)} \tag{J3}$$

Shrinking the contour to zero then shows that J is the energy made available for crack extension under elastic conditions. However, under conditions of plasticity, since W is not recoverable energy, J is <u>not</u> the energy made available for separation. Nevertheless, for the deformation theory of plasticity J remains a path-independent integral, subject to the other interpretations to follow. However, the above relationship to K allows the J-Integral to be used to evaluate K directly for various analytical and finite element numerical analyses.

EXAMPLE OF THE USE OF THE *J*-INTEGRAL IN EVALUATING *G* AND *K*

Consider the mixed boundary value problem for the configuration shown in **Fig. J.2a**. It shows a sheet stretched uniformly prior to clamping rigid, smooth parallel boundaries perpendicular to the stretch direction, and with a centrally introduced crack from one edge subsequent to clamping. The sheet is considered to be

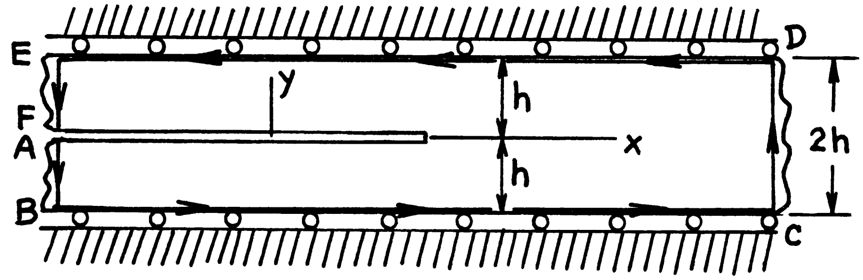


Fig. J.2a

elastic but not necessarily linear. This problem is analogous to the vertical periodic array of cracks problem of **page 12.1**. The application of the J-Integral follows by integrating on the contour shown. On segments $A-B$ and $E-F$, where all stresses are relieved, there are no tractions, nor any strain energy, W, so that portion of the integral gives zero. Further on segments $B-C$ and $D-E$, dy, T_x, and $\frac{\partial u_y}{\partial x}$ are zero so the integration there gives zero. On the final segment $C-D$, the tractions are zero but the strain energy, W, remains undisturbed by the crack and is constant with y.

Therefore, integration over the full contour, Γ, is reduced to

$$J = G = \int_{-h}^{h} W dy = 2Wh \quad \text{(nonlinear elastic)} \tag{J4}$$

However, if the sheet is linear elastic, then

$$J = G = \frac{\sigma^2}{E} h = \frac{K^2}{E} \quad \text{(linear elastic)} \tag{J5}$$

This otherwise difficult problem for finding G or K is made almost trivial by the J-Integral method. For such special problems, the method can be very useful, including use with numerical methods such as finite element methods into the elastic–plastic regime.

RELATIONSHIP OF J TO CRACK OPENING STRETCH, δ_T, FOR THE STRIP YIELD MODEL

The strip yield model, as discussed in **Section 30** and its solution pages, has a crack opening stretch at the crack tip that is directly related to J as follows. **Figure J.2b** shows any of the crack tip situations for all configurations. The shaded plastic zone is assumed to be perfectly plastic with a yield strength, σ_0, applied to

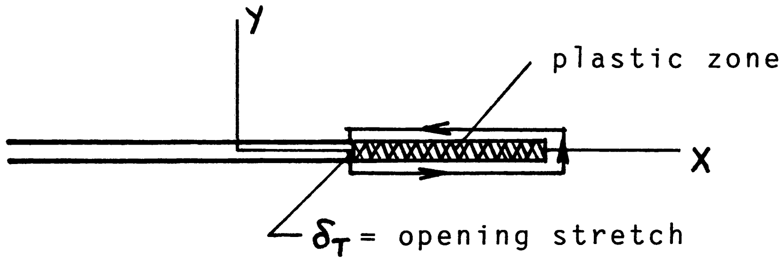


Fig. J.2b

the zone as a constant traction, $T_y = \sigma_0$, top and bottom. Then, since dy is zero, and ds is simply dx or $-dx$, integrating around the contour gives

$$J = -\sigma_0 \int \frac{\partial U_y}{\partial x} ds = \sigma_0 \delta_T \qquad \textbf{(J6)}$$

This simple result allows conversion to J for the strip yield results in Section 30. However, the idealization of strip yielding can simply be dropped to allow the yielding to spread, in which case it is still found that the relationship is given by

$$\delta_T = \gamma \frac{J}{\sigma_0} \qquad \textbf{(J7)}$$

where γ is a constant of the order of one, depending on plane stress or plane strain yield zone conditions and the hardening coefficient of the material. In any event, this J-Integral calculation has demonstrated the appropriate type of relationship in a direct fashion.

ALTERNATE "NONLINEAR COMPLIANCE" DEFINITION OF J

The preceding nonlinear elastic energy rate interpretation of J itself suggests further alternative forms for the definition of J for both elastic and elastic–plastic circumstances. Consider, a nonlinear elastic body containing a crack and loaded as shown in **Fig. J.3**. Suppose the body is loaded by a force, P, whose work-producing component of displacement is denoted δ. For elastic conditions the load displacement diagram might look as shown in **Fig. J.3** up to point A for constant crack length, a, where the value of J is desired. At that point, if the displacement is fixed, and the crack is advanced by an increment, da, the load will drop to point B with no further work of loading induced. Then at point B, with crack size now $a + da$, unloading will result in the curve from B to O. The region $O - A - B - O$ on that load displacement curve is the energy

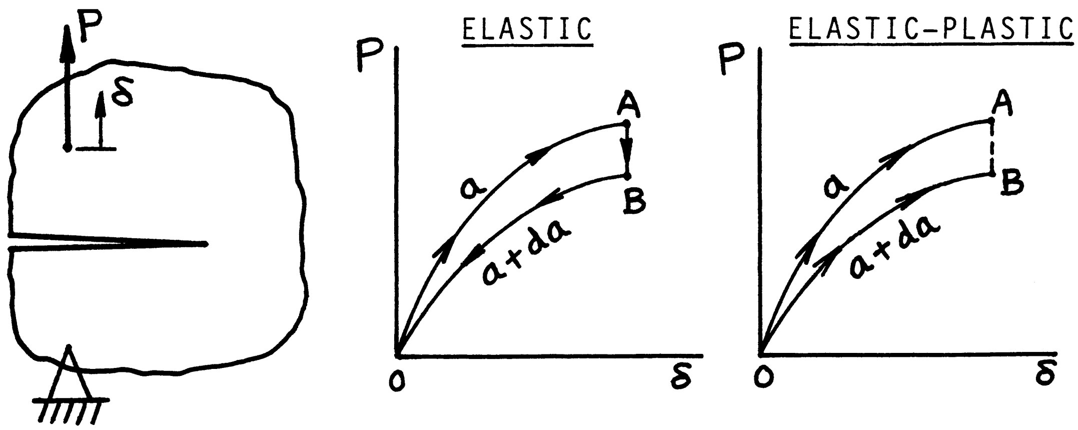


Fig. J.3

given up for the crack extension *da*, and is therefore equal to *Jda*. Considering that fixity of the displacement in the preceding argument would change the result only by a higher-order double differential, it can be seen that fixity does not influence the result. Therefore, *J* may be determined from

$$J = -\int_0^{\delta} \frac{\partial P}{\partial a} d\delta = \int_0^{P} \frac{\partial \delta}{\partial a} dP \qquad (\mathbf{J8})$$

These two integral forms in terms of load displacement relations are equally valid definitions of *J*, consistent with the original contour integral definition given by **(J1)**. Indeed, the elastic–plastic load displacement curves for two samples of crack sizes *a* and $a + da$ on the final load displacement curve in **Fig. J.3** would be identical to those in the nonlinear elastic case under deformation theory of plasticity assumptions, if the loading stress–strain curves of the elastic and elastic–plastic materials were identical. Therefore, these alternate definitions of *J*, by the above integrals, also equally apply under deformation theory restrictions for the evaluation of *J*.

These alternate definitions of *J* made possible the experimental technique used in the famous works of **Landes (1972)** and **Begley (1972)** for measurement of *J* under elastic–plastic conditions. Further, for purposes of computing or estimating *J* in practice, the above "nonlinear compliance" form of *J* leads to the following consideration. For linear–elastic plastic conditions the displacement can always be exactly divided into its elastic and plastic parts. That is to say:

$$\delta = \delta_{el} + \delta_{pl} \qquad (\mathbf{J9})$$

The second integral form then becomes

$$J = \int_0^{P} \frac{\partial \delta_{el}}{\partial a} dP + \int_0^{P} \frac{\partial \delta_{pl}}{\partial a} dP \qquad (\mathbf{J10})$$

Now the first of these two integrals is exactly the linear–elastic energy rate, G, and the second can be converted by analysis of load versus plastic displacement (plastic part only) to result in

$$J = J_{el} + J_{pl} = G - \int_0^{\delta_{pl}} \frac{\partial P}{\partial a} d\delta_{pl} \qquad \textbf{(J11)}$$

This form has a great advantage when used to estimate J under elastic–perfectly plastic conditions (no hardening), because for large plastic deformations it gives

$$J \cong G - \frac{\partial P_{\lim}}{\partial a} \delta_{pl} \qquad \textbf{(J12)}$$

where $P_{\lim}$ is the fully plastic limit load. Actually, since the first term, G, in this expression is small for large plastic deformation, the second term by itself is often an adequate estimate of J.

THE HUTCHINSON–RICE–ROSENGREN (HRR) CRACK TIP FIELD EQUATION INTERPRETATION OF *J*

Hutchinson (1968) and **Rice (1968b)** independently demonstrated that the near-tip stress and strain fields within the plastic region can be found for materials, if they are reasonably represented as purely power hardening well beyond their elastic range, which are the conditions near a crack tip in elastic–plastic materials even with moderate loading. A power-hardening material is represented by the stress–strain relation

$$\frac{\varepsilon}{\varepsilon_0} = \left(\frac{\sigma}{\sigma_0}\right)^n \qquad \textbf{(J13)}$$

For such a material, in the intense plastic enclave at a crack tip, the field equations have a dominant term (the only singular term) of the forms

$$\sigma_{ij} = \sigma_0 \left(\frac{J}{\sigma_0 \varepsilon_0 r}\right)^{\frac{1}{n+1}} \Sigma_{ij}(\theta, n) \qquad \textbf{(J14)}$$

$$\varepsilon_{ij} = \varepsilon_0 \left(\frac{J}{\sigma_0 \varepsilon_0 r}\right)^{\frac{n}{n+1}} E_{ij}(\theta, n) \qquad \textbf{(J15)}$$

Again these equations were developed assuming deformation theory of plasticity with small strains and rotations. They are therefore ‘‘valid’’ in a region surrounding a crack tip, away from the tip by an amount measured by the crack opening stretch, but near to the tip compared with planar body dimensions and well within the crack tip plastic zone. Therefore, J may be interpreted as the strength of the plastic field surrounding the crack tip, just as K has been interpreted as the strength of the elastic field for small-scale yielding conditions, However, the J field strength does not depend on the presence of small scale yielding conditions! Further, it is clear that, for any monotonicaly increasing true stress–strain behavior, the power law assumed is not necessary to have J as the relevant intensity parameter. Indeed, J is the parameter that reflects the effects of body configuration and loading on the surrounding crack tip plastic field.

J-CONTROLLED CRACK GROWTH

In various applications, it is desirable to use a parameter such as J to measure the effects of loading and configuration on crack tip deformation and stressing. To justify using J, the assumptions of deformation theory of plasticity must reasonably prevail in the surrounding crack tip field. The "no unloading" and "proportional straining" assumptions are of concern if the crack is extending, da, as loading or imposed deformation, as reflected by dJ, is occurring. To consider the proportionality of straining, **Hutchinson (1979)** formed the differential of strain $d\varepsilon_{ij}$ to compare it with the preceding strain field equation. The result was of the form

$$d\varepsilon_{ij} = \varepsilon_{ij}\left(\frac{n}{n+1}\right)\frac{dJ}{J} + \left(\frac{\partial\varepsilon_{ij}}{\partial r}\cdot\frac{\partial r}{\partial a} + \frac{\partial\varepsilon_{ij}}{\partial\theta}\cdot\frac{\partial\theta}{\partial a}\right)da \qquad \textbf{(J16)}$$

where the parentheses of the second term, if multiplied by a characteristic planar dimension, is of the order of the first term excluding the dJ/J. However, the second term results in an r, θ distribution of strain that is nonproportional to the original strains. Hence proportional straining only occurs for situations where the complete first term dominates the second. This is the case if

$$\frac{dJ}{J} \gg \frac{da}{b} \qquad \textbf{(J17)}$$

where b is the relevant characteristic dimension for the field, which is normally the remaining uncracked ligament ahead of the crack. On this basis the conditions for valid "J - controlled crack growth" are stated as

$$\omega = \frac{dJ}{da}\cdot\frac{b}{J} \gg 1 \quad (\text{and } \Delta a_{total} \ll b) \qquad \textbf{(J18)}$$

For example, the "validity" of $J - R$ curve data is limited to portions of the curve where its current slope $\frac{dJ}{da}$ divided by current J is large enough for the specimen size, as measured by b to keep $\omega \gg 1$. Valid conditions for crack growth must also apply in any proper structural application.

FURTHER DIMENSIONAL CONSIDERATION OF THE HRR FIELD EQUATIONS

Note that the preceding HRR field equations imply that J is characterized by stress times strain times a characteristic length. In the analysis of a cracked body with a single characteristic dimension (the crack size) subject to a uniform remote field of stress (and thereby strain), the form of the applied J must be

$$J = C\sigma\varepsilon a \qquad \textbf{(J19)}$$

where C is a constant for the particular configuration (an infinite plate with a crack of length $2a$ perpendicular to the stress direction, or a semi-infinite plate with an edge crack or an internal circular crack or semi-circular surface flaw, etc.). For example, the familiar Griffith configuration gives

$$J = G = \frac{\pi\sigma^2 a}{E} = \pi\sigma\varepsilon a \qquad \textbf{(J20)}$$

Now, for example, applying this result to a material that is represented appropriately by the Ramberg-Osgood relation

$$\frac{\varepsilon}{\varepsilon_0} = \frac{\sigma}{\sigma_0} + \alpha\left(\frac{\sigma}{\sigma_0}\right)^n \qquad \textbf{(J21)}$$

then the relationship for J for this particular configuration becomes

$$J = \pi\left\{\left(\frac{\sigma}{\sigma_0}\right)^2 + \alpha\left(\frac{\sigma}{\sigma_0}\right)^{n+1}\right\}\sigma_0\varepsilon_0 a \tag{J22}$$

This is an excellent form for approximating J when the applied stress, σ, exceeds σ_0 for the Griffith configuration, and for others with the single characteristic dimension of the crack, by appropriately adjusting the constant π in this expression. The first change is for the constant adjustment in the elastic solution, for example, the π should be replaced by $^4/_\pi$ for the penny-shaped circular crack. Second, the coefficient, α, should also be increased because of hardening, approximately by the factor $(1 + 0.26n)$ to accommodate fully adjustments for the nonlinear material behavior term according to **Paris (1983a)**.

A reasonable approximation for configurations with more than one characteristic dimension is to use the linear elastic configuration correction factors for stress levels, σ, at or not far above σ_0. The final form for such approximations uses both the above stress correction bracket and a geometry correction bracket as well, in the form

$$J = \sigma_0\varepsilon_0 a\{\mathit{Stress}\}[\mathit{Geometry}] \tag{J23}$$

The separated nondimensional brackets are convenient in applying Tearing Instability Theory for the stability of growing cracks using R-curve concepts, as well as other applications. (It should be noted that the EPRI Elastic-Plastic Fracture Analysis Handbook method is based on similar approximations from dimensional considerations.)

RICE'S ANALYSIS FOR PURE BENDING OF A REMAINING LIGAMENT

For a semi-infinite plate with a semi-infinite crack approaching the free edge leaving a ligament subject to pure bending, as shown in **Fig. J.4**, dimensional analysis leads to a very useful result. For this configuration, the relative angle change, θ, between the applied moments (per unit thickness), M, must depend only on that moment, M, and the only characteristic dimension, b, the ligament size, and the stress–strain properties of the material (whose dimensions are those of stress, $^F/_{L^2}$, and strain, nondimensional). Therefore, M and b must appear in a combination to give $^F/_{L^2}$ to allow θ to be nondimensional. Formally stated, θ must be

$$\theta = f\left(\frac{M}{b^2},\ \textit{stress-strain}\right) \tag{J24}$$

This functional form is true for both elastic and plastic behavior. The form may be inverted to give

$$M = b^2 F(\theta) \tag{J25}$$

where $F(\ )$ includes the materials properties (constant). Considering that $da = -db$, then this result may be put into the alternative nonlinear compliance form for the J-Integral to give

$$J = -\int_0^\theta \left.\frac{\partial M}{\partial a}\right|_\theta d\theta = +\int_0^\theta 2bF(\theta)d\theta = \frac{2}{b}\int_0^\theta M d\theta \tag{J26}$$

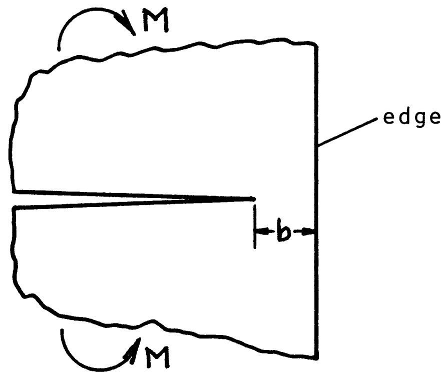


Fig. J.4

where the crack size, b, remains constant and the final integral of $Md\theta$ is the work done in loading. This result is independent of the actual elastic–plastic stress–strain relation as long as no crack growth occurs. Again, for pure bending, it is

$$J = \frac{2}{b}(Work) \quad \text{(no crack growth)} \tag{J27}$$

where ($Work$) is per unit thickness of plate. For other configurations with a single characteristic dimension similar simple forms are possible, (see **Rice 1973**).

Now, when crack growth is occurring under J-controlled growth conditions so that deformation theory of plasticity applies, J may be considered to be a function of deformation, θ, and crack size, a, (or b): $J = J(\theta, a)$ and an increment in J must be computed by

$$dJ = \frac{\partial J}{\partial \theta} d\theta + \frac{\partial J}{\partial a} da \tag{J28}$$

Forming dJ for pure bending of a ligament from this result and **(J26)** gives

$$dJ = \frac{2}{b} d(Work) - \frac{J}{b} da \quad \text{(for } J\text{-controlled growth)} \tag{J29}$$

Consequently, J may be computed as an increment-by-increment accumulation under J-controlled crack growth conditions. For other configurations similar forms may be written for dJ, including those with more than one characteristic dimension (see **Ernst 1979**). However, the principles have been adequately demonstrated here.

As a final example, showing the results of the preceding dimensional arguments, the formula developed by these means for the determination of J for ASTM standard Compact Specimen tests is

$$J + \Delta J = \left[J + \frac{\eta}{b}(\Delta\ Work)\right]\left[1 - \frac{\gamma}{b}\Delta a\right] \tag{J30}$$

where

$$\begin{aligned} \eta &= 2 + 0.522\left({}^{b}\!/_{W}\right) \\ \gamma &= 1 + 0.76\left({}^{b}\!/_{W}\right) \end{aligned} \tag{J31}$$

and the current values of each parameter here should be used in evaluating the next increment, ΔJ. Here the 2 in η, and the 1 in γ are limiting factors agreeing with the preceding analysis where W is large compared to b, tending toward pure bending of the remaining ligament.

The result that the work in pure bending of a remaining ligament is directly related to J, the strength of the crack tip field of deformation, undoubtedly gives a sound basis for the use of bending tests, such as the Charpy impact test, to evaluate fracture toughness. Indeed, that result, which does not rely on the particular stress–strain curve of the material but only on the consequent work of the loading, is surprising in its generality. Another example of a surprising result follows here to demonstrate the unique qualities of the J-Integral.

J FOR THE DOUBLE CANTILEVER BEAM CONFIGURATION

The configuration of a double cantilever beam is shown in **Fig. J.5**, with opposing concentrated loads

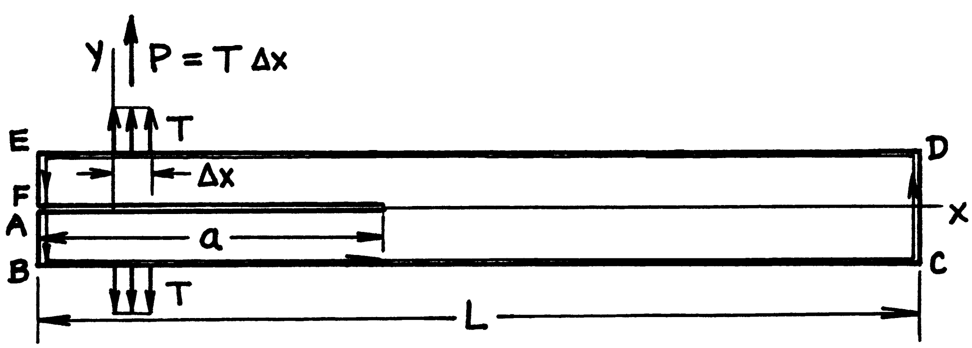


Figure J.5

modeled by uniform tractions over short segments (**A.J. Paris 1988**). The original contour J-Integral form, **(J1)**, will be used to evaluate J over the contour $A - B - \cdots - F$. Note that Wdy is zero over the whole contour and tractions are zero everywhere except at the load points where $T_y = T$. Therefore the integral becomes

$$J = \oint \left(Wdy - T_i \frac{\partial u_i}{\partial x} ds \right) = -\int_0^{\Delta x} T \frac{\partial u_y}{\partial x} dx + \int_{\Delta x}^{0} T \frac{\partial u_y}{\partial x} dx \qquad (\mathbf{J32})$$

where the items in the final integrands are constant. The partial derivatives of the displacements at the load points are the relative rotations of the arms with respect to each other, totaling θ, which gives

$$J = P\theta \qquad (\mathbf{J33})$$

This result gives J in terms of the load, P, and relative arm rotations, θ, independent of the elastic–plastic stress–strain properties, in terms of the final values of load and rotation, without requiring values during loading. This is a most amazingly simple result, showing the power of the J-Integral in analyzing the effects of loading on the intensity of the stress–strain singularity at a crack tip.

ELASTIC–PERFECTLY PLASTIC ESTIMATES OF *J*

For elastic–perfectly plastic behavior (no hardening) of the load-displacement relationship for a crack body, J can be estimated with good accuracy in terms of linear-elastic results and limit load behavior, as follows. The alternate nonlinear compliance form of J is given by **(J11)** and **(J12)**

$$J = G - \int_0^{\delta_{pl}} \frac{\partial P}{\partial a} d\delta_{pl} = G - \frac{\partial P_{\lim}}{\partial a} \delta_{pl} \tag{J34}$$

where the rate of change of limit load, $P_{\lim}$, with crack size is often known (always negative) or easy to compute. Especially for large plastic deformations, this is an accurate and practical way to determine J. In addition to many other potential applications, this method provides a direct way to evaluate J in bending of beams, tubes (pipes), and the like where plastic hinges form at cracked sections in statically determinate or indeterminate structures. Notice that this method can be easily and consistently incorporated into "plastic design" approaches, including collapse mechanism analysis. It is also effective in analyses of crack stability in beams under J-controlled growth of cracks (see **Paris 1983b**).

CRACK STABILITY UNDER *J*-CONTROLLED GROWTH CONDITIONS

Tearing instability theory, as initially developed by **Paris (1979)** and others, such as **Hutchinson (1979)** identifying conditions for applying J-controlled growth, allows evaluations of crack stability for linear and nonlinear elastic and elastic–plastic conditions where deformation theory of plasticity applies, that is, for all J-controlled growth conditions. It argues that equilibrium of a growing crack is expressed by the first derivative of energy, U, with respect to crack size, which is expressed as

$$J = \frac{dU}{da} = R \quad \text{(equilibrium)} \tag{J35}$$

where R is the material's resistance to crack growth in terms of energy or work required per unit increase in crack area. R is frequently presented as an "R-curve" of a material in terms of resistance, R, for a given crack extension, Δa, from its initial size. The J here is considered to be the applied J reflecting the loading and configuration of the deformed body. Crack stability then depends on the second derivative of energy, or

$$\frac{dJ}{da} = \frac{d^2U}{da^2} (< \text{or} \geq) \frac{dR}{da} \quad (< \text{stable}; \ \geq \ \text{unstable}) \tag{J36}$$

where $\frac{dR}{da}$ is the slope of a J-Integral R-curve.

Defining the nondimensional parameters

$$T_{appl} = \frac{dJ}{da} \cdot \frac{E}{\sigma_0^2} \quad \text{and} \quad T_{mat} = \frac{dR}{da} \cdot \frac{E}{\sigma_0^2} \tag{J37}$$

stability is judged by

$$T_{appl} \ (< \text{or} \geq) \ T_{mat} \quad (< \ \text{stable}; \geq \text{unstable}) \tag{J38}$$

It is convenient to plot diagrams of T vs. J to analyze stability continuously as J increases with loading or deformation. These diagrams identify first instability somewhat more conveniently than techniques employing using tangency with R-curves (see **Paris 1983a**).

APPLICATION OF C^* (TIME DEPENDENT J) TO CREEP CRACK GROWTH

Begley (1976) showed that a modified J-Integral, denoted C^*, replacing strain and displacement with their time derivatives, is a controlling parameter for certain creep crack growth behaviors. In such cases the integral is defined as

$$C^* = \oint \left(W' dy - T_i \frac{\partial \dot{u}_i}{\partial x} ds \right) \tag{J39}$$

where

$$W' = \int \sigma_{ij} d\dot{\varepsilon}_{ij} \tag{J40}$$

and the dot over them indicates time derivatives of ϵ and u. Later discussions of this time rate-modified J-Integral have been provided by **Riedel (1980, 1989)**, in which its further relevance to applications has been given.

APPLICATION OF ΔJ TO FATIGUE CRACK GROWTH

Dowling (1976) suggested computing the range of J or ΔJ added during a load cycle in fatigue and correlated that parameter with the cyclic rate of crack growth. The ΔJ in this work was defined in cyclic plastic bending of a remaining ligament as

$$\Delta J = \frac{2}{b} \Delta(Work) \tag{J41}$$

where Δ *Work* was defined as the area under the crack-opened portion of the loading cycle. This parameter correlated fatigue crack growth rates under cyclic plastic conditions, and also related these rates to other fatigue cracking rates for tests in the small-scale yielding regime by

$$\Delta K = \sqrt{E' \Delta J} \tag{J42}$$

Although the assumptions of deformation theory and thus J normally do not allow unloading, the success of this method may be taken to imply no effects of the prior unloading cycles, that is, no history dependence. Indeed, these last two examples of applying J to time-dependent and cyclic loaded subcritical crack growth demonstrate the power of J as a relevant analysis parameter.

In conclusion J has been demonstrated here to be a powerful analytical tool for a wide range of applications, from providing a method to evaluate K and G under elastic conditions to nonlinear elastic and elastic–plastic applications of correlating cracking behavior under static, time-dependent, and even cyclic deformation and loading. It generalizes analysis under a broader scope, including as a special case, linear-elastic method employing K and G. Its arguments and assumptions go beyond but are completely consistent with linear-elastic fracture mechanics methods.

APPENDIX K

ELASTO-PLASTIC PURE SHEAR STRESS-STRAIN ANALYSIS (MODE III)

This appendix presents a method of developing "exact solutions" to elastic-perfectly plastic problems in pure shear for the stresses and strains in both the elastic and plastic regions. Since no such analytical method is available for any other (Mode I or II) crack problems, this Mode III method is given to illustrate the nature of plasticity in this rather special case as it may aid intuitively in more general problems. The method to be presented was fully developed in **Rice (1966, 1968c, 1984)** and similar problems were previously investigated by F. A. McClintock in the 1950s.

Pure shear (or anti-plane shear) results from problems where the deformations can be described in or are restricted to the following form of the induced displacements:

$$u(x, y, z) = 0, \quad v(x, y, z) = 0, \quad w(x, y, z) = w(x, y)$$

This form for the displacements leads to strains of the form

$$\varepsilon_x = \varepsilon_y = \varepsilon_z = 0$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = 0, \quad \gamma_{yz} = \frac{\partial w}{\partial y}, \quad \gamma_{xz} = \frac{\partial w}{\partial x}$$

Differentiating each of these two nonzero strains with respect to the opposite variable and equating leads to the compatibility equation

$$\frac{\partial \gamma_{xz}}{\partial y} - \frac{\partial \gamma_{yz}}{\partial x} = 0 \qquad (\mathbf{K1})$$

The absence of the three extensional strains as well as the xy shear strain implies the absence of the following stress components:

$$\sigma_x = \sigma_y = \sigma_z = \tau_{xy} = 0$$

Therefore the only nontrivial equilibrium equation is

$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} = 0 \tag{K2}$$

Equations (K1) and (K2) are usually solved directly for elastic problems, also making use of the stress-strain relations

$$\gamma_{xz} = \frac{\tau_{xz}}{G} \quad \text{and} \quad \gamma_{yz} = \frac{\tau_{yz}}{G} \tag{K3}$$

The solution to this set of equations, **Eqs. (K1), (K2), and (K3)**, would normally take the form

$$\tau_{xz} = \tau_{xz}\ (x, y) \quad \text{and} \quad \tau_{yz} = \tau_{yz}\ (x, y) \tag{K4}$$

However, for elastic–plastic problems, advantages occur upon inverting the variables and solving the problem on the stress plane, the τ_{xz} vs. τ_{yz} plane, rather than on the physical plane, the x vs. y plane. The solutions will then involve mapping points on the physical plane onto the stress plane, which is sometimes called the "Hodograph Method" in other applications.

INVERTING TO THE STRESS PLANE

The stress equations. **Eq. (K4)**, can be inverted to express the coordinates as functions of the stresses, or

$$x = x\ (\tau_{xz},\ \tau_{yz}) \quad \text{and} \quad y = y\ (\tau_{xz},\ \tau_{yz}) \tag{K5}$$

Taking the differentials of **Eqs. (K5)** and then substituting for the differentials of stress from **Eqs. (K4)** leads to two equations, which, upon noting that dx and dy are independent, become four equations. Performing some direct algebraic manipulation one can then show that **Eq. (K2)** becomes

$$\frac{\partial x}{\partial \tau_{xz}} + \frac{\partial y}{\partial \tau_{yz}} = 0 \tag{K6}$$

Similarly, **Eqs. (K1) and (K3)** may be combined and inverted to give

$$\frac{\partial x}{\partial \tau_{yz}} - \frac{\partial y}{\partial \tau_{xz}} = 0 \tag{K7}$$

which applies only to the elastic region of any problem.

STRESS FUNCTION SOLUTION FOR THE ELASTIC REGION

Figure K.1 shows the stress plane where the circle represents the yield surface for homogeneous isotropic yielding and points within the circle are the elastic region on this plane. For the solution within this elastic region, a stress function, Φ, is elected where

$$\Phi = \Phi(\tau_{xz},\ \tau_{yz})$$

and

$$x = \frac{\partial \Phi}{\partial \tau_{yz}}, \quad y = -\frac{\partial \Phi}{\partial \tau_{xz}} \tag{K8}$$

so that the equilibrium equation, **Eq. (K6)**, is automatically satisfied and the elastic region compatibility equation, **Eq. (K7)**, becomes

$$\frac{\partial^2 \Phi}{\partial \tau_{xz}^2} + \frac{\partial^2 \Phi}{\partial \tau_{yz}^2} = \nabla^2 \Phi = 0 \tag{K9}$$

which is the well-known Laplace's Equation.

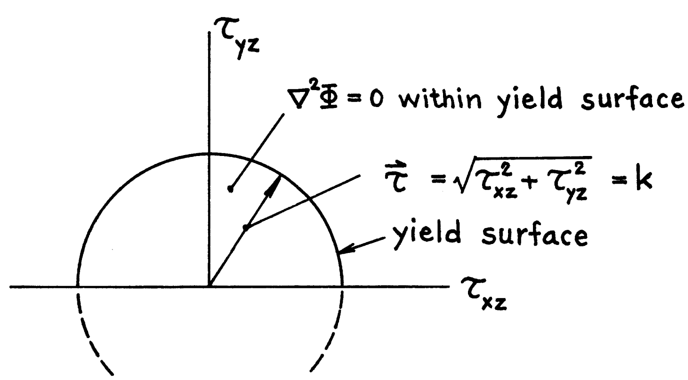
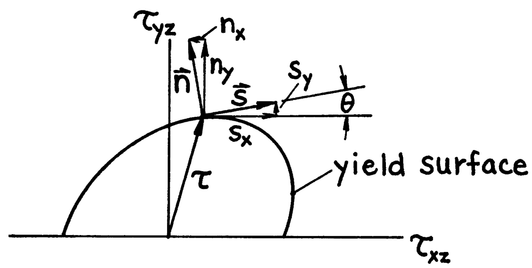


Fig. K.1

BOUNDARY CONDITIONS

The boundary conditions for the solution of **Eq. (K9)** in the elastic region of the stress plane for typical crack problems, with stress boundary conditions on external surfaces, are dictated by the mapping of those external surfaces from the physical plane onto the stress plane, as will be illustrated. However, the boundary conditions for Φ at the elastic–plastic boundary are more subtle. **Figure K.2** shows an anisotropic yield surface (other than circular) with *ds* defined as an increment of length along that surface at the point indicated by the stress vector, τ. Further, the normal unit vector to the yield surface, $\vec{n}$, and the tangent unit vector, $\vec{s}$, and their components are shown on the stress plane. In addition the corresponding crack tip is shown on the physical plane in **Fig. K.2** with the plastic zone at its tip. The plastic slips must begin at the singularity of the crack tip forming a staircase of slips emanating at that tip. For elastic–perfectly plastic material (no hardening) the emphasized slip at an angle θ must correspond to the point on the yield surface with the corresponding angle θ because, as is well known in plasticity theory, plastic strains must be normal to the yield surface.

Stress Plane



Physical Plane

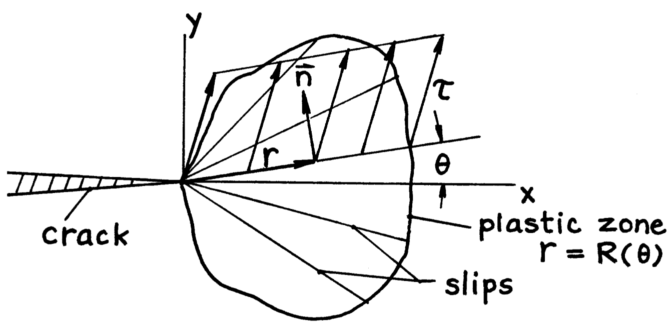


Fig. K.2

From conditions on the yield surface it is noted that

$$s_x = n_y = \frac{\partial \tau_{xz}}{\partial s}, \quad s_y = -n_x = \frac{\partial \tau_{yz}}{\partial s}$$
$$\text{and } \vec{n} = n_x\vec{i} + n_y\vec{j} \tag{K10}$$

Also on the physical plane

$$\vec{r} = x\vec{i} + y\vec{j} \tag{K11}$$

Now the rate of change of the stress function along the yield surface may be written as

$$\frac{d\Phi}{ds} = \frac{\partial \Phi}{\partial \tau_{xz}}\frac{\partial \tau_{xz}}{\partial s} + \frac{\partial \Phi}{\partial \tau_{yz}}\frac{\partial \tau_{yz}}{\partial s}$$

which upon relevant substitutions from **Eqs. (K8), (K10), and (K11)** give

$$\frac{d\Phi}{ds} = -(xn_x + yn_y) = -\vec{r}\cdot\vec{n} = 0 \tag{K12}$$

since the vectors $\vec{r}$ and $\vec{n}$ are perpendicular. Therefore the boundary condition along the yield surface on the stress plane is that Φ is a constant, which may be arbitrarily chosen to be zero, thus

$$\Phi = \underline{\text{ constant} = 0} \text{ (on the yield surface)} \tag{K13}$$

In addition, the radial coordinate on the physical plane from the crack tip to the elastic–plastic boundary may be denoted

$$R(\theta) = \sqrt{x_\theta^2 + y_\theta^2} = \sqrt{\left(-\frac{\partial \Phi}{\partial \tau_{xz}}\right)^2 + \left(\frac{\partial \Phi}{\partial \tau_{yz}}\right)^2}$$

or

$$R(\theta) = \sqrt{\left(\frac{\partial \Phi}{\partial n}\right)^2 + \left(\frac{\partial \Phi}{\partial s}(=0)\right)^2} = \frac{\partial \Phi}{\partial n} \tag{K14}$$

From this expression, the physical radius of the plastic zone is determined along any radial slip line by the gradient of the stress function on the stress plane.

Let us further invoke Neuber's observation that at a crack tip the stress singularity times the strain singularity of the field is always $1/r$. For no hardening, the stress is a nonsingular, constant, τ, along radial lines within the plastic zone. Therefore the strain along these radial lines must be determined by

$$\gamma_{\theta z} = \gamma_{r=R}\left(\frac{R(\theta)}{r}\right) = \frac{\partial w}{r \partial \theta}$$

$$\gamma_{rz} = \frac{\partial w}{\partial r} = 0$$

Consequently, the stresses and strains are completely known within the plastic zone once the stress function, Φ, is determined for the adjacent elastic region. The discussion now proceeds to the determination of Φ.

MAPPING INDIVIDUAL PROBLEMS AND FULL BOUNDARY CONDITION DETERMINATION

Figure K.3 shows the example of a half plane with a crack of length, a, perpendicular to its edge. The crack is along the x-axis with its tip at the origin making the edge parallel to the y-axis. The body is loaded with uniform shear stress, $\tau_{yz} = \tau_0$, at $y = \pm\infty$.

Physical Plane

Stress Plane

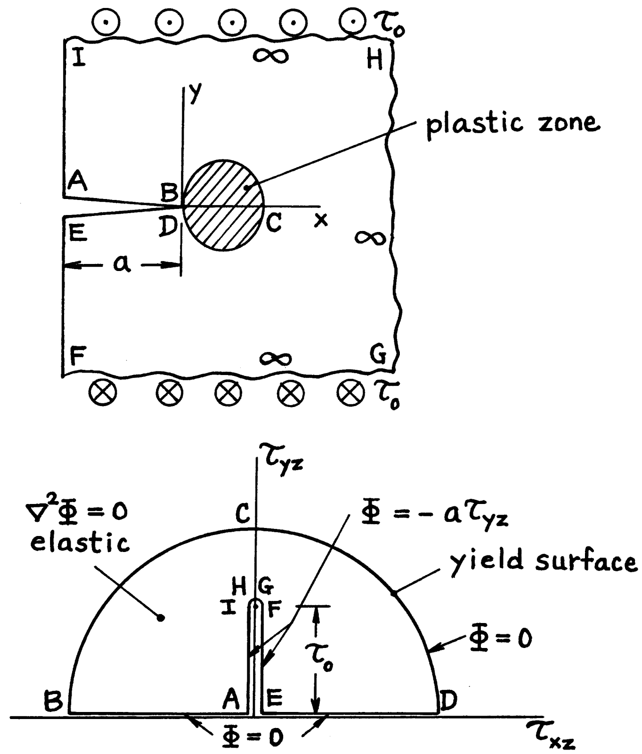


Fig. K.3

In addition, the stress plane is shown with a circular yield surface corresponding to isotropic yield strength, k, or the arc $\tau^2 = \tau_{xz}^2 + \tau_{yz}^2 = k^2$. To complete the mapping, successive points around the boundary of the elastic region on the physical plane are labeled $A, B, C, ..., I$, which correspond to similarly labeled points on the stress plane. For example, all along the upper crack surface is labeled AB, and $\tau_{yz} = 0$ as a load free boundary condition. At the corner at A it is entirely stress free, or $\tau_{xz} = 0$, whereas the magnitude of this stress component gradually increases along AB, meeting the yield surface, $\tau_{xz} = -k$, so that B is at the origin on the physical plane but at the yield surface on the stress plane. Note that where the plastic zone crosses the x-axis remote from the origin, labeled C on the physical plane, because of symmetry, $\tau_{xz} = 0$ so that on the stress plane $\tau_{yz} = +k$. Similar to AB on the stress plane, DE occupies the $\tau_{xz}(= +)$ axis from the origin to the yield surface. Further, for all boundaries at infinity on the physical plane corresponding to $FGHI$ the stresses are $\tau_{yz} = +\tau_0$ and $\tau_{xz} = 0$, that is, all at the same point on the stress plane. It is also noted that points along EF and IA on the physical plane are on the vertical axis on the stress plane. The map is then completed and the elastic region is within the boundary, as described on the stress plane. Within that boundary the stress function, Φ, obeys **Eq. (K9)** or $\nabla^2\Phi = 0$ as indicated. It remains to express the boundary conditions on Φ on the stress plane. They are:

a.) On the elastic–plastic boundary, BCD, **Eq. (K13)** specifies

$$\Phi_{BCD} = 0$$

b.) On AB and DE it is noted that

$$y = 0 = -\frac{\partial \Phi}{\partial \tau_{xz}} \quad \text{along} \quad \tau_{yz} = 0$$

Therefore integrating gives

$$\Phi_{AB} = \Phi_{DE} = \quad \text{constants} \quad = 0$$

where the zero values are required to avoid a discontinuity in the boundary condition as it passes through B and D on the stress plane.

c.) On EF and IA it is noted that

$$x = -a = \frac{\partial \Phi}{\partial \tau_{yz}} \quad \text{along} \quad \tau_{xz} = 0$$

Again integrating this result gives

$$\Phi_{IA} = \Phi_{CF} = -a\tau_{yz} + 0$$

where the constants of integration are again both taken to be zero to avoid a discontinuity passing through A and C on the stress plane.

These boundary conditions on Φ are noted on the stress plane map in **Fig. K.3** and it is left to solve for it within these boundaries by using Laplace's **Eq. (K9)**. The final solution is left undone, as the object of this example is to illustrate the setting up of the problem for solution. The exact analytical solution to a similar problem shall now be given because of its special significance to small-scale yielding fracture mechanics assumptions.

THE COMPLETE ANALYTICAL SOLUTION FOR SMALL-SCALE YIELDING

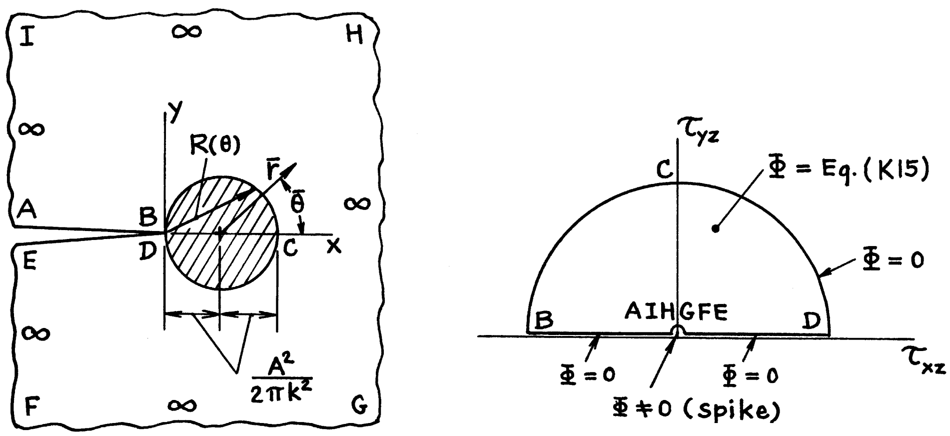
A significant special case of the preceding example is that for which the free edge is removed to infinity giving an infinite region on the physical plane with a semi-infinite crack with its tip at the origin, as shown in **Fig. K.4**. The loading on the boundaries as shown on **Fig. K.3** must approach zero but not without leaving significant finite stresses and strains, causing the usual singular effects at the crack tip. The physical plane and mapping to the stress plane are shown in **Fig. K.4** for this case. Note that the boundary condition for Φ exhibits singular behavior at the origin of the stress plane.

The stress function for this problem is

$$\Phi(\tau_{xz}, \tau_{yz}) = \frac{A^2}{2\pi}\left[\frac{1}{\tau_{xz}^2 + \tau_{yz}^2} - \frac{1}{k^2}\right]\tau_{yz} \tag{K15}$$

Note that this stress function satisfies Laplace's Equation and gives zero values on the yield surface and horizontal axis of the stress plane except at the origin where a singular spike in value occurs. Differentiating this stress function using **Eqs. (K8)**, the elastic–plastic boundary on the physical plane is described by

$$y^2 + \left(x - \frac{A^2}{2\pi k^2}\right)^2 = \left(\frac{A^2}{2\pi k^2}\right)^2 \tag{K16}$$



Physical Plane Stress Plane

Figure K.4

which is a circle directly ahead of and just touched by the crack tip. Further, it is found that larger circles with the same center on the physical plane correspond to semi-circles centered at the origin within the elastic region on the stress plane. For mapping corresponding points from the physical plane to the stress plane on these circles, the angular measure from their centers differs by the factor 2. Consequently, stresses in the elastic region may be most simply expressed in terms of new coordinates, $\bar{r}$, $\bar{\theta}$, measured on the physical plane from the center of the plastic zone given by **Eq. (K16)**, and they are

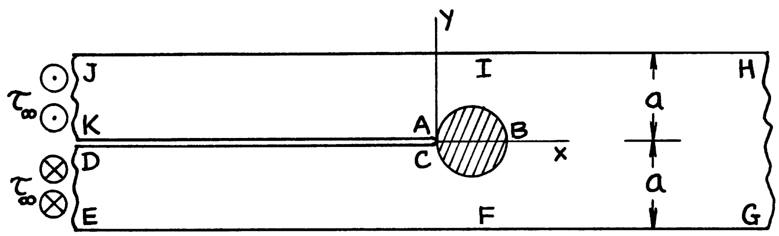
$$\tau_{xz} = -\frac{A}{\sqrt{2\pi\bar{r}}}\sin\frac{\bar{\theta}}{2}$$

$$\tau_{yz} = \frac{A}{\sqrt{2\pi\bar{r}}}\cos\frac{\bar{\theta}}{2} \qquad (\mathbf{K17})$$

Comparing these equations with **Eq. (3)** in Part I of this handbook, it is noted that exactly the same form of elastic stress field is found for the purely elastic analysis except that here the coordinates are taken at the center of the plastic zone instead of at the origin. That means that the plastic zone embeds itself within the same elastic stress field as that for the purely elastic case, except that the "effective (or equivalent) elastic crack length" is the physical crack length plus the plastic zone radius. These same "plastic zone corrections" to crack size, as suggested by **Paris 1957** and later developed by **Irwin (1960b)**, have been used without full analytical justification for many years for Mode I linear-elastic fracture mechanics applications. Although this analytic justification is shown only for a Mode III case here, it adds significant credibility for the concept applied to other modes. Indeed, it shows that a small-scale plastic zone embeds itself within the elastic crack tip stress field without significantly disturbing the form of that field but with an effective crack size including half the plastic zone.

EXAMPLES OF OTHER MAPS

Physical Plane



Stress Plane

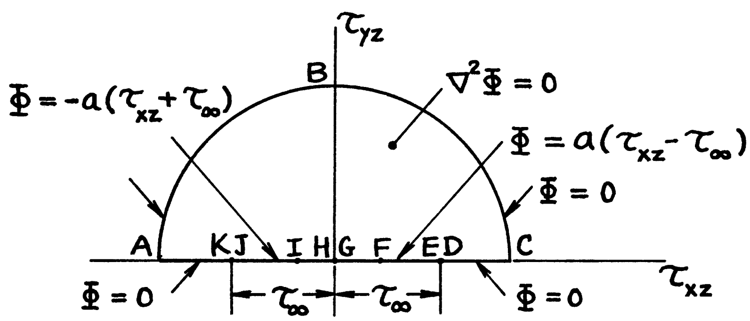


Fig. K.5

For the example shown in **Fig. K.5**, it is noted that
a.) On *CD* and *AK*:

$$y = 0 = -\frac{\partial \Phi}{\partial \tau_{xz}} \quad \text{or} \quad \Phi_{CD} = \Phi_{AK} = C_1 = 0$$

b.) On *HIJ* and *EFG*:

$$y = \pm a = -\frac{\partial \Phi}{\partial \tau_{xz}} \quad \text{or} \quad \begin{array}{l} \Phi_{EFG} = a(\tau_{xz} - \tau_\infty) \\ \Phi_{HIJ} = -a(\tau_{xz} + \tau_\infty) \end{array}$$

with simultaneous points *DE* and *KJ* at $\tau_{yz} = 0$, $\tau_{xz} = \pm\tau_\infty$.